

DESIGN THINKING & INNOVATION REPORT



**VASIREDDY VENKATADRI
INSTITUTE OF TECHNOLOGY**

TEAM-14

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CERTIFICATE

This is to certify that the project titled “**Emergency Ambulance Tracking System**” is a Bonafide record of work done by **K. Navya Sri, P. Deepthi Sri, N. Nishitha, P. Vyshnavi, K. Bhanu Sri**, under the guidance of **Mr. S. Kishore Babu, Associate Professor** in partial fulfillment of the requirement for the award of credits to **Design Thinking & Innovation** -course of Bachelor of Technology in Artificial Intelligence and Data Science (AI & DS), JNTUK during the academic year 2025- 26.

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ACKNOWLEDGEMENT

We express our sincere thanks wherever it is due

We express our sincere thanks to the **Chairman**, Vasireddy Venkatadri Institute of Technology, **Sri Vasireddy Vidya Sagar** for providing us well equipped infrastructure and environment.

We thank **Dr. Y. Mallikarjuna Reddy, Principal**, Vasireddy Venkatadri Institute of Technology, Nambur, for providing us the resources for carrying out the project.

We express our sincere thanks to **Dr. K. Giri Babu, Dean of Academics**, for providing support and stimulating environment for developing the project.

Our sincere thanks to **Dr. T. Sudhir, Head of the Department, Department of AI & DS**, for his co-operation and guidance which helped us to make our project successful and complete in all aspects.

We also express our sincere thanks and are grateful to our guide **Mr. Saggurthi Kishore Babu, Associate Professor, Department of AI & DS**, for motivating us to make our project successful and fully complete. We are grateful for his precious guidance and suggestions.

We also place our floral gratitude to all other teaching staff and lab technicians for their constant support and advice throughout the project.

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Problem Statement

In emergency medical situations, every second counts. Patients and bystanders across India and globally face a critical challenge — they cannot track the real-time location of dispatched ambulances, leading to severe delays in emergency response. The absence of a transparent, GPS-enabled ambulance tracking system results in increased patient mortality, miscommunication between hospitals and dispatchers, and inefficient fleet management. There is an urgent need for a comprehensive Emergency Ambulance Tracking System that provides live GPS tracking, intelligent route optimization, real-time hospital bed availability, and seamless communication between patients, dispatchers, drivers, and medical staff — ultimately reducing response time and saving lives.

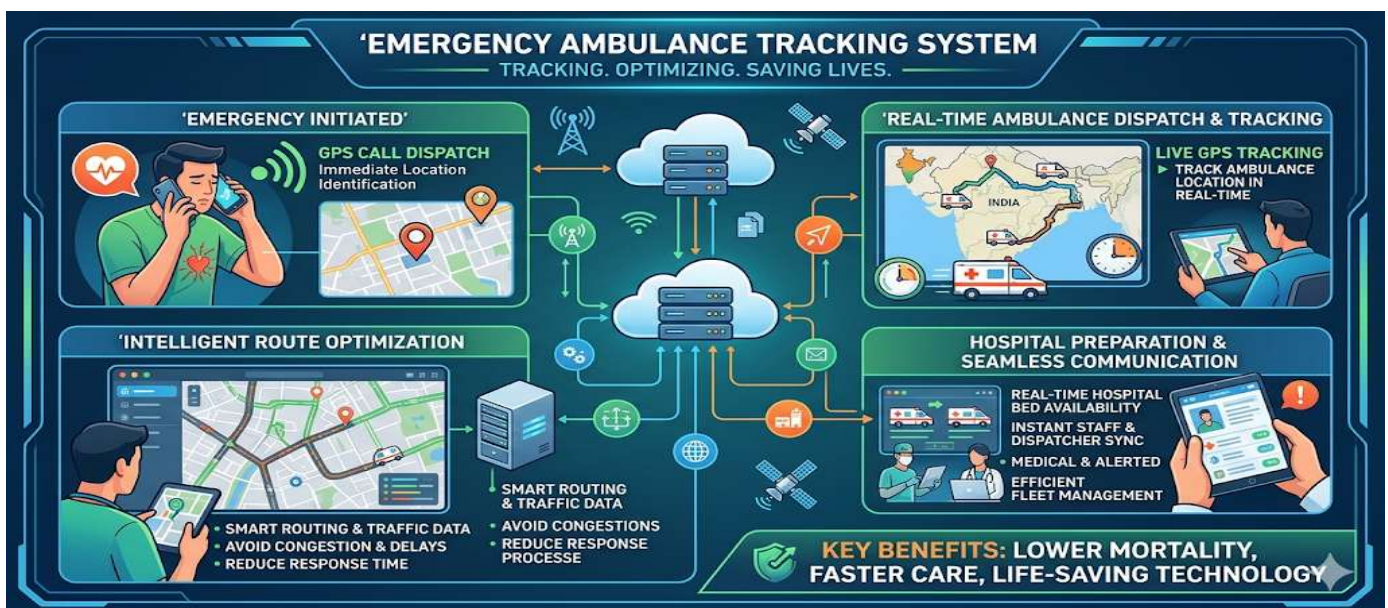


Figure Caption:Emergency Ambulance Tracking System illustrating GPS-based dispatch, real-time tracking and communication (Source: Generated using Gemini AI)

Phases Of Design Thinking

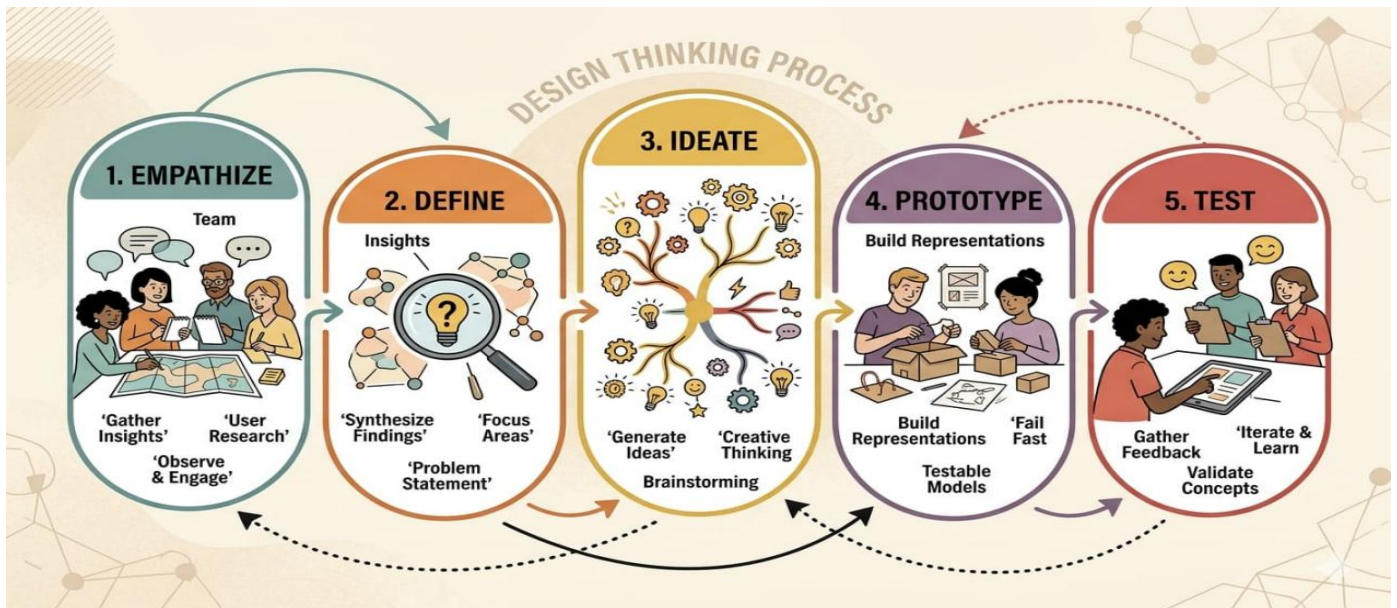


Figure caption: The five stages of the Design Thinking process: Empathize, Define, Ideate, Prototype, and Test, showing the iterative flow between user research and concept validation (Source: Generated using Gemini AI).

1. Empathy

The first phase of the Design Thinking process is Empathy, where the goal is to understand the users' needs, emotions, motivations, and challenges. In the context of our Emergency Ambulance Tracking System, the focus is on understanding the pain points faced by patients, families, ambulance drivers, hospital dispatchers, and emergency responders during critical medical emergencies.

1.1 Brainstorming (Explore & Exploit)



Figure 1.1: Idea Board Showcasing brainstorming concepts and solutions for helping students discover career paths aligned with their academic streams (Source: Generated using Gemini AI)

Objective:

The purpose of this brainstorming session was to understand the current challenges in ambulance dispatching and emergency response, and to explore how a real-time tracking application can effectively reduce response times and save lives.

Explore: Understanding the Problem Space

We initiated the brainstorming session by identifying broad questions and challenges related to emergency ambulance tracking:

- Are you aware of ambulance services available in your area?

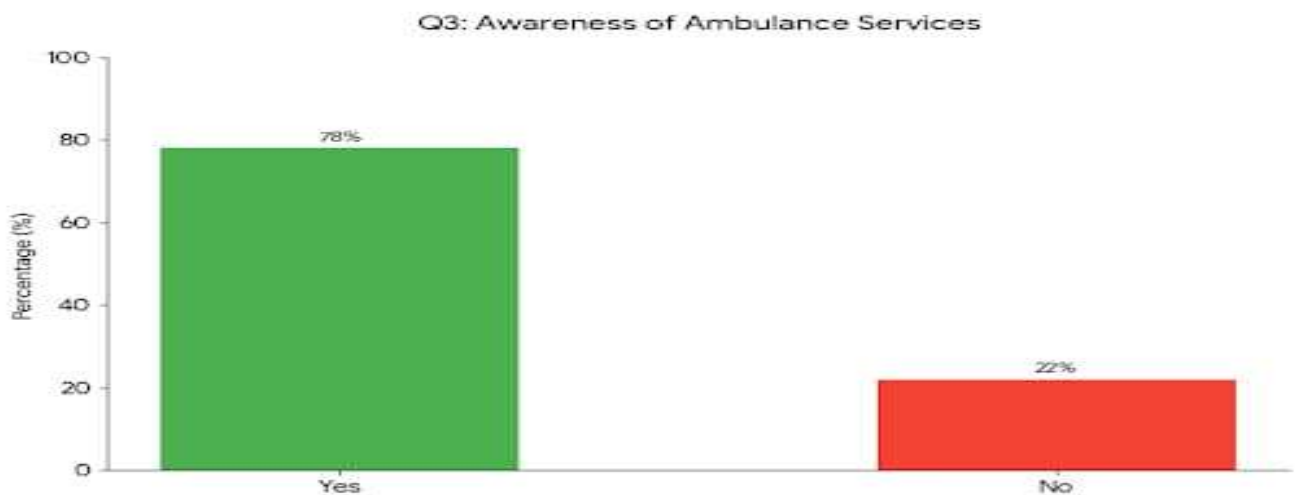
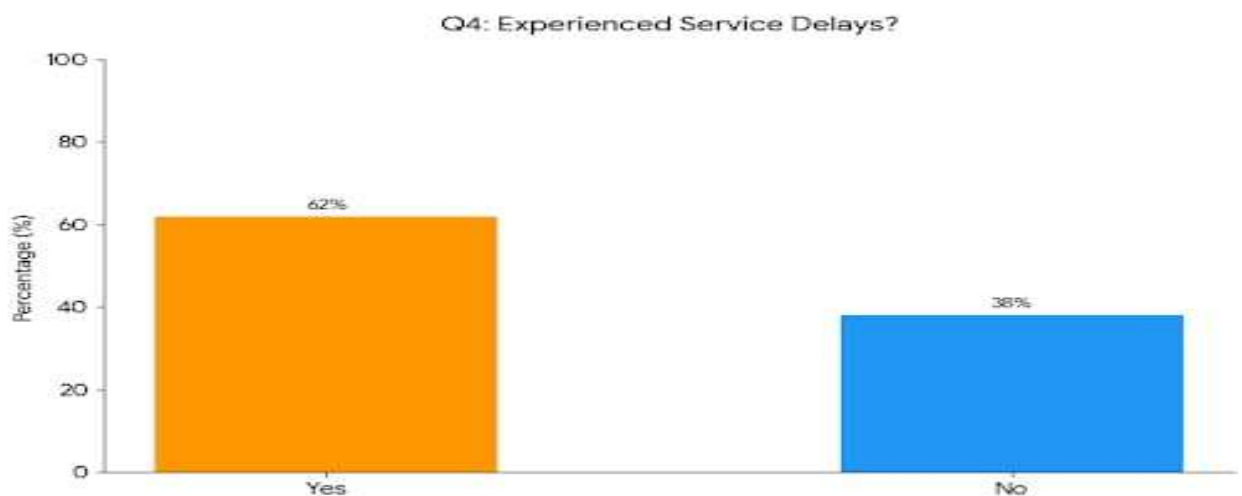


Figure Caption: Distribution of local ambulance service awareness among survey respondents

(Source: Generated using Type Form)

- Have you ever experienced delay in ambulance services?



- **Figure Caption:**Percentage of respondents who have previously encountered delays in emergency medical transit(**Source:** Generated using Type Form)

During this phase, we encouraged all team members to share unfiltered thoughts to expand the scope of understanding. Sticky notes, mind maps, and whiteboard sketches were used to record and organize ideas.

Key Findings from Exploration:

- Most patients and families have no visibility into ambulance dispatch status or ETA.
- Drivers rely on basic GPS apps not optimized for emergency navigation or traffic preemption.
- Dispatchers manually coordinate using phone calls, which introduces delays and errors.
- There is no integrated system linking hospitals, ambulances, and dispatch centers in real-time.
- Patients with critical conditions (cardiac arrests, trauma) need faster, data-driven triage routing.

Exploit: Identifying Opportunity Areas

Real-Time GPS Ambulance Tracking:

Provide a live map showing the ambulance's exact position to patients, family, and hospitals.

Smart Route Optimization:

Use traffic data and emergency signal preemption algorithms to identify the fastest ambulance path.

Hospital Readiness Integration:

Connect ambulance systems with hospital dashboards showing live bed availability and specialist status.

Dispatcher Control Panel:

Offer a centralized dashboard for dispatchers to assign, monitor, and reroute ambulances in real-time.

Patient SOS & Feedback:

Enable one-tap SOS from mobile, with automated status notifications sent to family members.

1.2 List of Stakeholders/Customers

Understanding who will interact with the Emergency Ambulance Tracking System and who will be affected by its implementation is crucial for building a user-centric solution.

Primary Stakeholders (Direct Users):

Patients and Bystanders:

- Need immediate access to ambulance location and ETA during emergencies.
- Require one-tap SOS and real-time status notifications.
- Are the core end-users of the patient-facing mobile app.

Ambulance Drivers and Paramedics:

- Need optimized navigation, hospital readiness data, and patient vitals during transport.
- Require a reliable in-vehicle interface integrated with dispatch.

Secondary Stakeholders:

Emergency Dispatchers:

- Use the platform to assign ambulances, monitor fleets, and communicate in real-time.
- May provide operational feedback to improve dispatch algorithms.

Hospital Emergency Staff:

- Need advance notification of incoming patients and their condition.
- Require bed occupancy and specialist availability dashboards.

Family Members:

- Want live tracking of ambulance to know if help is en route.
- Need SMS/app notifications on ambulance arrival times.

Tertiary Stakeholders:

Government Health Authorities:

- Interested in response time analytics and fleet performance data.
- Can leverage data for policy and infrastructure planning.

Insurance Companies:

- May use verified response time records for claims processing.

Technology Partners and Telecom Providers:

- Provide GPS, network, and cloud infrastructure for the system.

1.3 Customer Interaction

To develop an effective and user-centered Emergency Ambulance Tracking System, we directly interacted with potential users to gather firsthand insights. This interaction phase focused on understanding real problems faced during emergency medical situations.

Purpose of Customer Interaction:

- To gather deep insights into user behavior, fears, and expectations during emergencies.
- To validate the problem identified during the brainstorming phase.
- To discover how patients, drivers, and dispatchers currently experience the ambulance system.

Method of Interaction:

We used a combination of semi-structured interviews and online surveys to connect with a diverse group of users — patients' families, ambulance drivers, ER nurses, and dispatchers.

Target Participants:

- 20 patients' family members who have used ambulance services
- 8 ambulance drivers and paramedics
- 5 hospital emergency dispatchers
- 6 ER nurses and doctors
- 10 general public respondents

Sample Interview Questions:

1. Have you ever faced a situation where you couldn't track an ambulance you called?

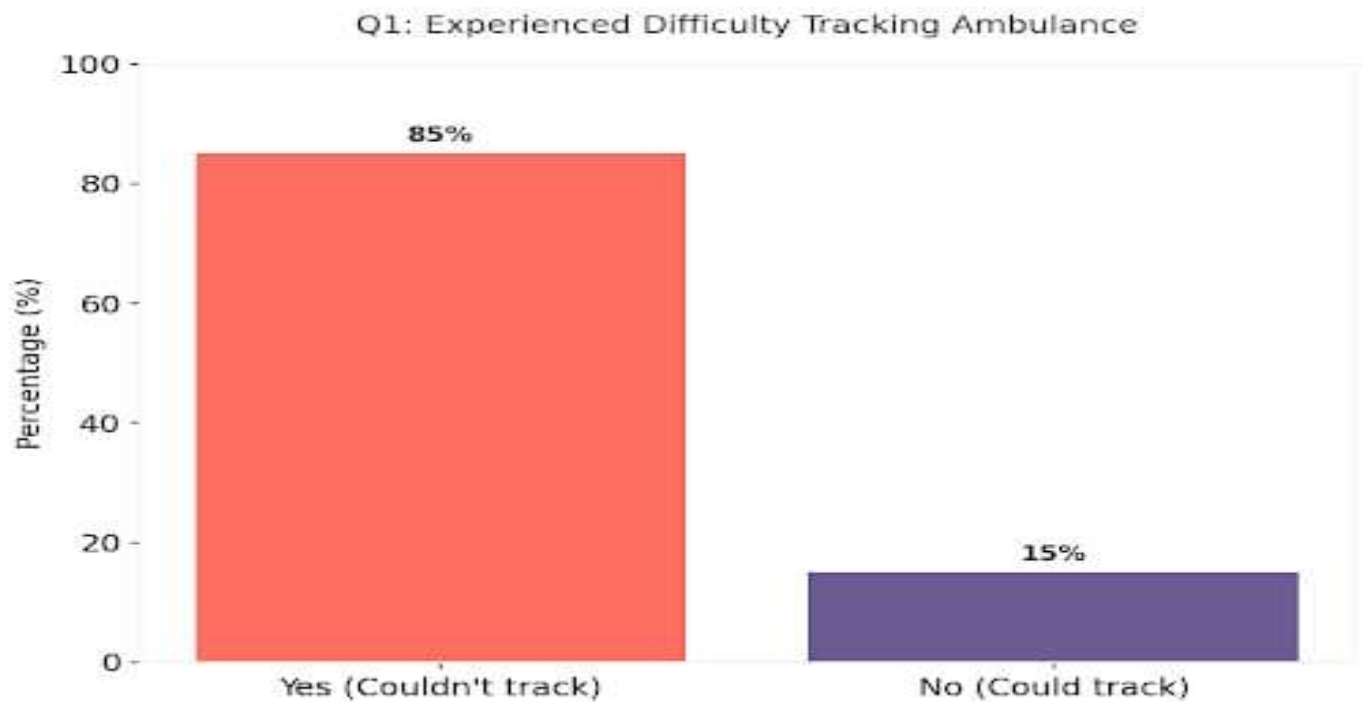


Figure 1.3.1: Percentage of respondents who experienced difficulty tracking an ambulance. (Source: Generated using Gemini AI)

2. What was the biggest difficulty during the emergency response process?

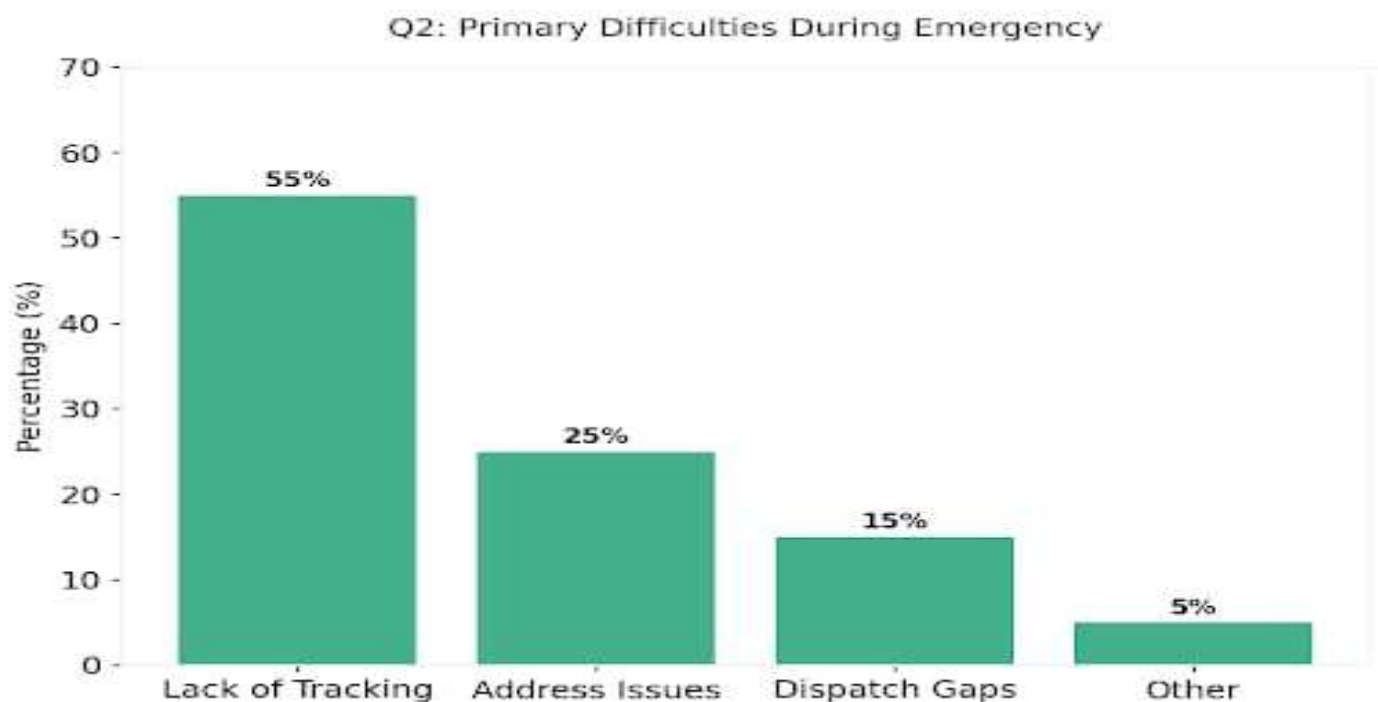


Figure 1.3.1: Distribution of primary difficulties faced by families during the emergency response process. (Source: Generated using Gemini AI).

3. Do you know the average ambulance response time in your city?

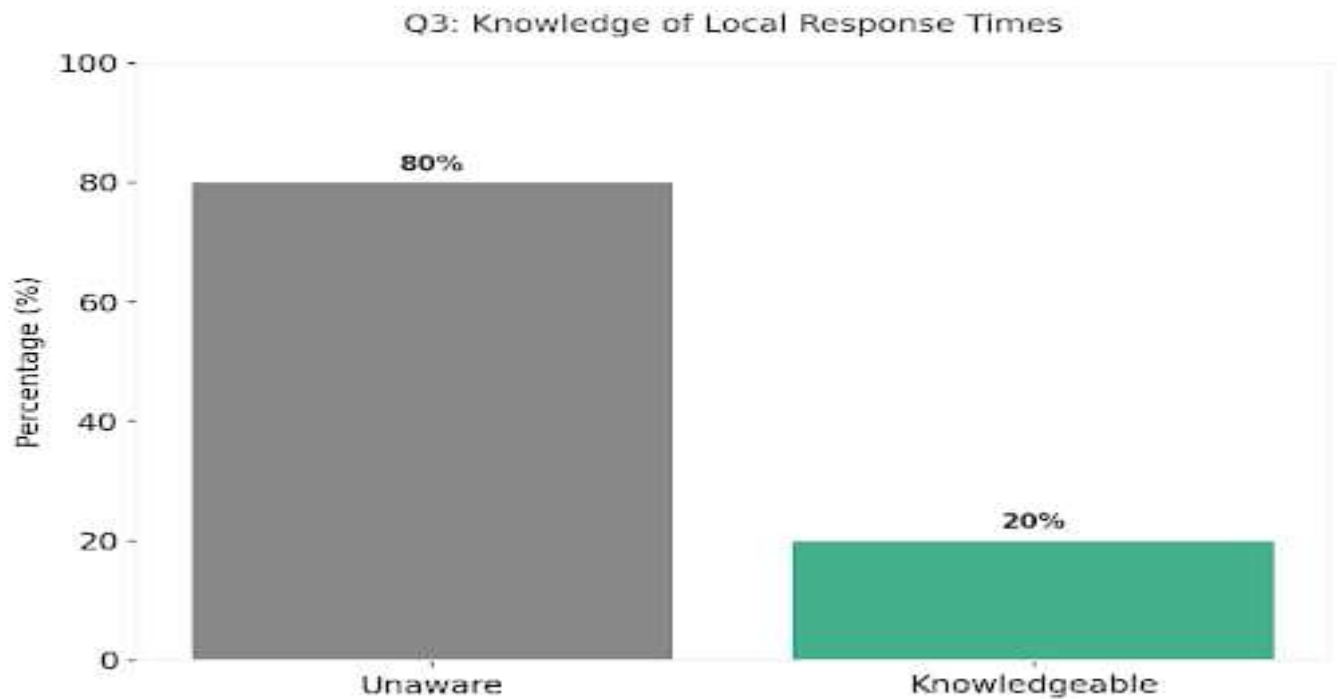


Figure 1.3.1: User awareness levels regarding average ambulance response times in their city. (Source: Generated using Gemini AI).

4. Would you use an app that lets you track a live ambulance and receive ETA updates?

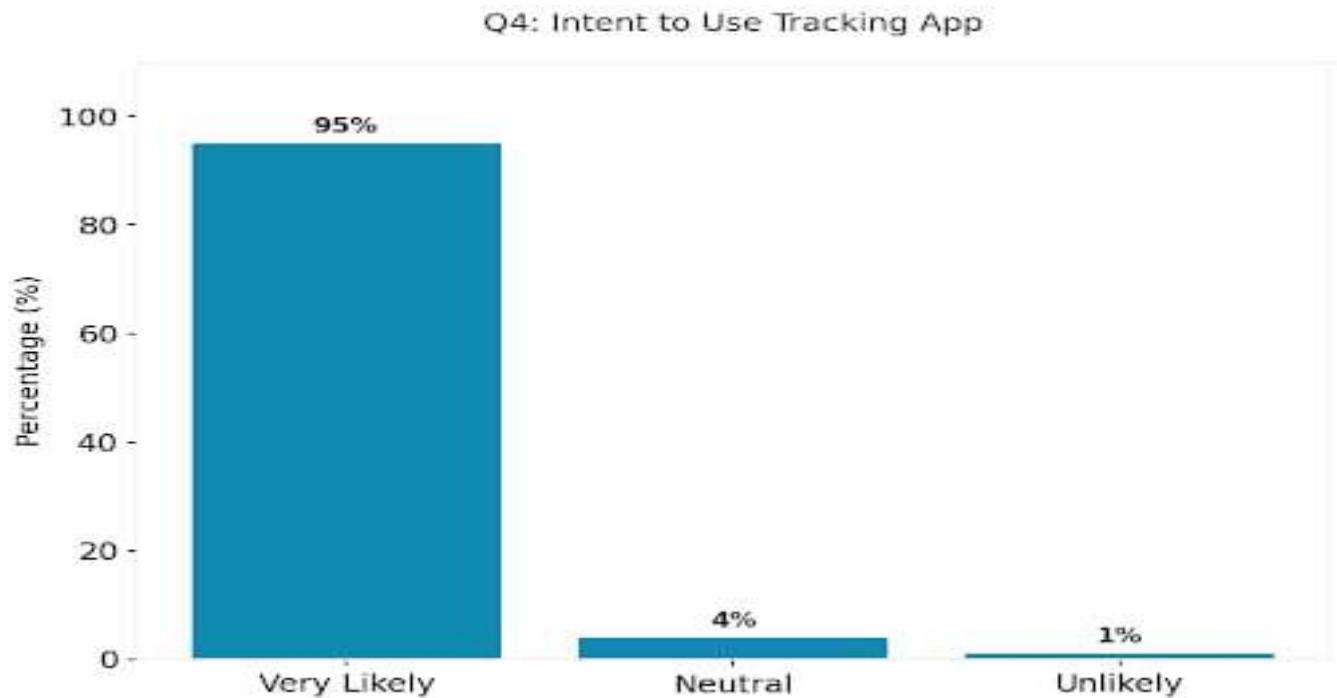


Figure 1.3.1: Stated intent of respondents to adopt and use a live ambulance tracking application. (Source: Generated using Gemini AI).

5. What features would make an ambulance tracking app most useful during a crisis?

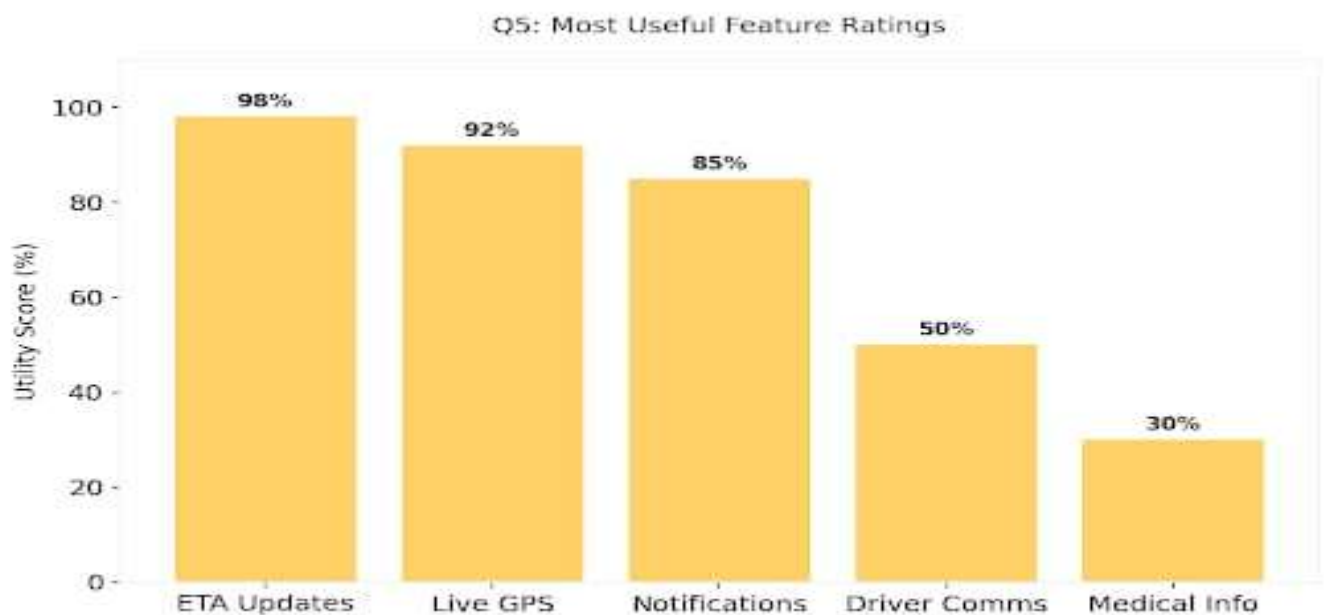


Figure 1.3.1: Ranking of the most useful features as identified by potential users. (Source: Generated using Gemini AI)

Key Insights from Patients and Families:

- Many families experience extreme anxiety because they have no idea where the ambulance is after calling.
- Several reported instances where the ambulance reached the wrong address due to poor communication.
- There is overwhelming demand for a live tracking app with push notifications during emergencies.

"We called the ambulance and waited for 40 minutes with no update. I didn't know if it was coming or lost." — Family member, cardiac emergency

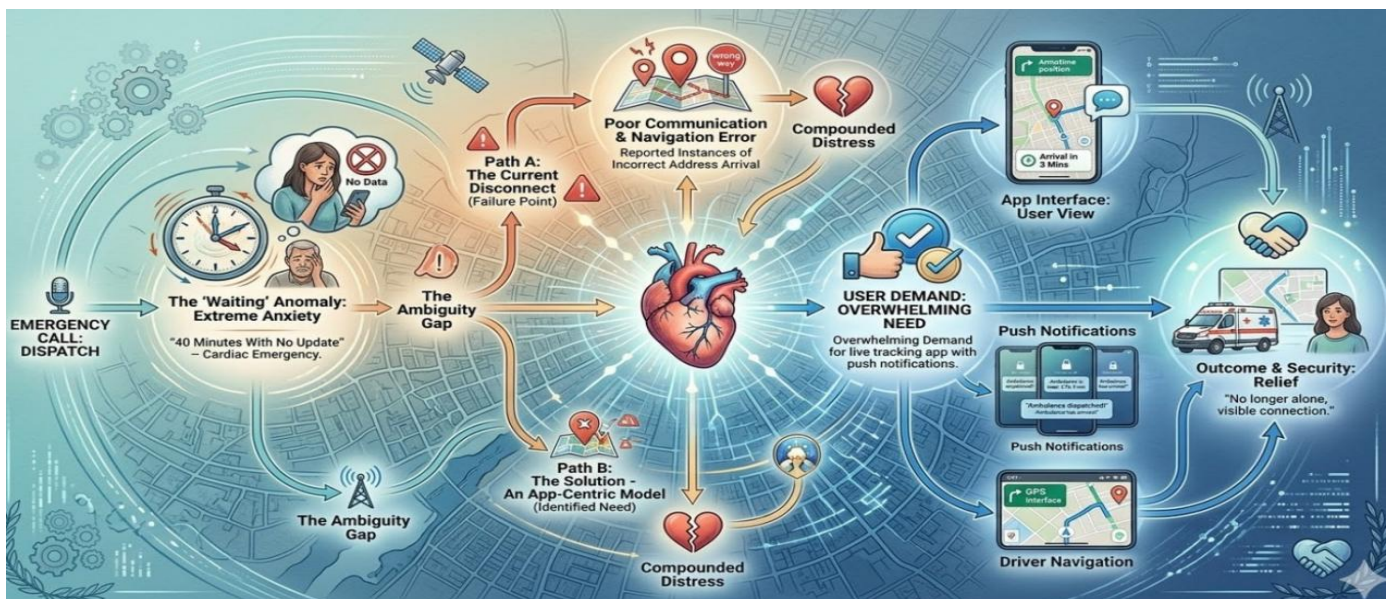


Figure 1.3: Service Design Map for Ambulance Tracking Solution (Source: Generated using Gemini AI)

Key Insights from Dispatchers and Drivers:

- Dispatchers use manual phone-based coordination, leading to errors in ambulance assignment.
- Drivers reported frequently getting stuck in traffic with no emergency route preemption.
- Drivers support a system that gives hospital readiness data before arrival.

"If I could see which hospital has a free trauma bay before I arrive, I could save at least 10 minutes." — Ambulance Paramedic, City Hospital

Key Insights from Hospital Staff:

- Hospitals often receive patients without any advance preparation time.
- Emergency rooms struggle with surge management due to lack of real-time ambulance ETAs.

1.4 Tools and Strategies Used

To deeply understand the users' mindset and challenges, we employed multiple empathy tools. These helped us capture both emotional and practical aspects of the emergency response experience.

Empathy Maps — Visualizing Patient Family Emotions

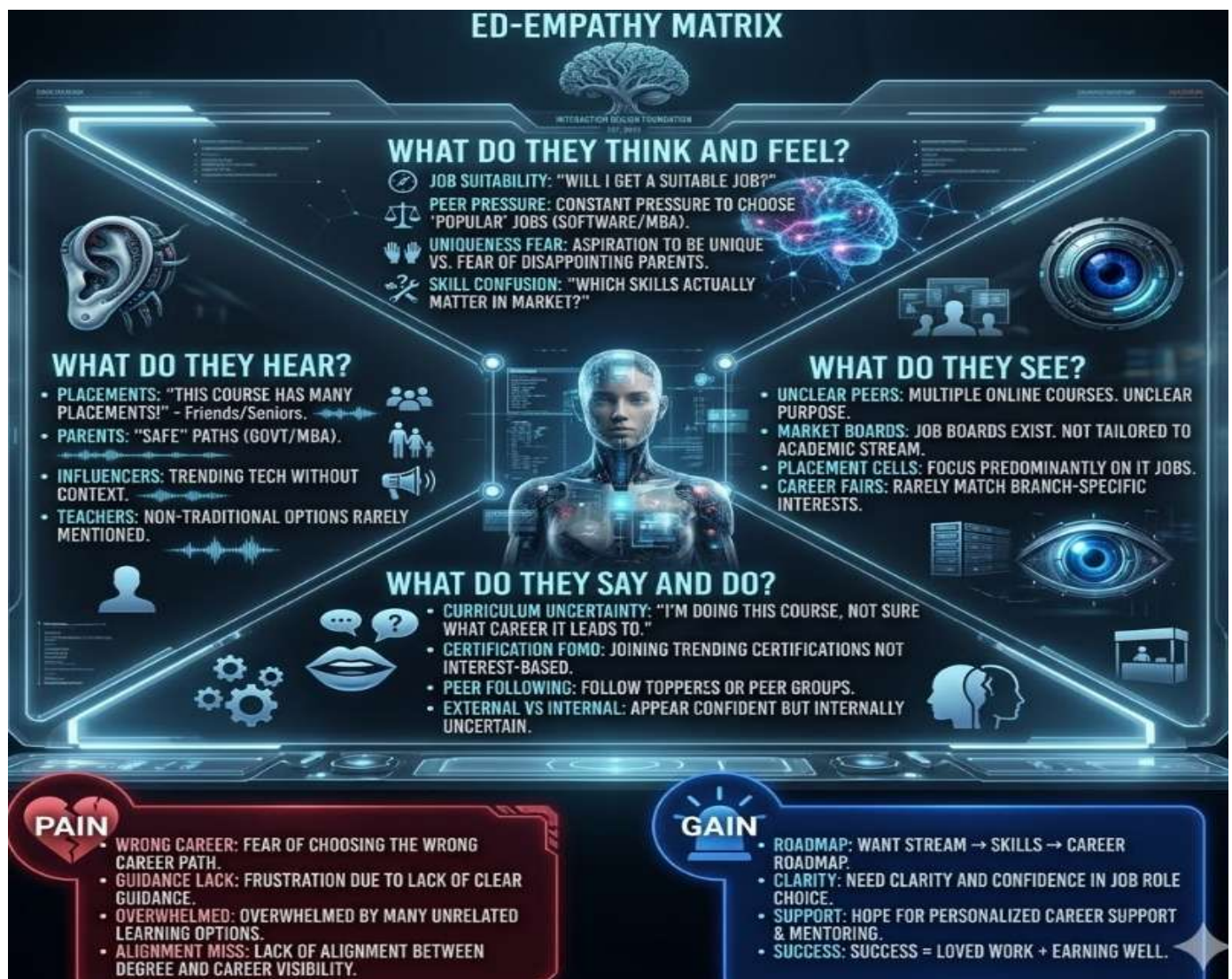


Figure 1.4: Empathy map detailing the psychological and environmental factors affecting career and skill decision-making. (Source: Generated using Gemini AI)

We created empathy maps to get a holistic view of what affected families say, think, do, and feel when waiting for an ambulance during an emergency. These maps helped us visualize deep-seated panic, helplessness, and the information void that currently exists.

Sample Empathy Map: "Ramesh — A 45-Year-Old Whose Father Had a Cardiac Arrest"

SAYS	THINKS
"The ambulance took 25 minutes. My father almost didn't make it."	"Will it reach in time? What if traffic blocks the route?"
"I couldn't track where the ambulance was during the entire wait."	"Is there a better system that can give me real-time updates?"
DOES	FEELS
Calls the ambulance dispatch repeatedly to check arrival status.	Panicked and helpless during the wait.
Tries to guide ambulance via phone call in heavy traffic.	Frustrated by the lack of transparent communication.
HEARS	SEES
Dispatch says: "We have dispatched an ambulance." No further update.	Crowded roads with no way to know if ambulance is nearby.
Neighbors say: "Just call 108 and hope for the best."	News reports of patients dying due to delayed ambulance response.

Insight Gained:

Families like Ramesh's need live location transparency, estimated arrival times, and proactive status updates. The system must eliminate the information void that causes helplessness during critical moments.

Surveys & Questionnaires — Broad User Data Collection

To reach a wider audience, we designed and distributed a survey using Google Forms. It consisted of 18 multiple-choice and short-answer questions targeting:

- Awareness of ambulance response times
- Experience with tracking or communication during emergencies
- Interest in a real-time ambulance tracking app
- Preferred features for an emergency response application

Participants:

- 150+ respondents from urban and semi-urban areas
- 10 healthcare professionals for expert validation

Key Survey Results:

- 88% of respondents said they had no visibility into ambulance location after calling for one.
- 76% reported experiencing anxiety or panic during the wait due to lack of information.
- 69% said poor routing and traffic caused the ambulance to arrive later than expected.
- 94% said they would definitely use a real-time ambulance tracking app if it were available.

Stakeholder Maps:



Figure 1.4.5: Stakeholder Engagement Matrix and Connectivity Hub. (Source: Generated using Gemini AI).

Feedback from Stakeholders:

Based on the sources, feedback from stakeholders was collected during the Empathy and Test phases through semi-structured interviews, surveys, and usability observations specific to the Emergency Ambulance Tracking System.

Key Insights from Patients and Families:

- Families are deeply distressed by the absence of real-time ambulance location data.
- Many have experienced misdirected ambulances due to inadequate GPS and address systems.
- There is a strong demand for an easy-to-use SOS and tracking solution.

Key Insights from Dispatchers and Drivers:

- Dispatchers rely heavily on manual processes which are error-prone and time-consuming.
- Drivers face constant routing challenges without dynamic navigation tied to emergency corridors.

Key Insights from Hospital Staff:

- Hospitals need advance preparation time and real-time incoming patient data.
- Integration between ambulance fleets and hospital systems would improve patient outcomes significantly.

2. Define

After gaining deep insights from the Empathy phase, the next step is to clearly define the core problems faced by users. This phase involves crafting Point-of-View (POV) statements that structure the problem from each user's perspective.

2.1 Point-of-View (POV) Statements

A POV statement combines user, needs, and insights into a clear and concise problem definition.

Ramesh — A 45-Year-Old Patient Family Member

User: Ramesh, whose father suffered a cardiac arrest

Need: Needs real-time GPS tracking of the ambulance and accurate ETA updates

Insight: Ramesh is overwhelmed by anxiety and has no information channel to track the ambulance

POV Statement:

Ramesh needs a transparent and real-time ambulance tracking solution with push notification updates because he feels helpless and terrified during the critical waiting period after calling for emergency help.

Priya — An Ambulance Dispatcher

User: Priya, a dispatcher at a city emergency call center

Need: Needs a centralized digital dashboard to assign and monitor multiple ambulances simultaneously

Insight: She currently manages all assignments via phone calls, causing delays and coordination failures

POV Statement:

Priya needs an intelligent, real-time dispatch management platform because her current manual process leads to errors, delays, and increased risk to patient lives.

Raju — An Ambulance Driver

User: Raju, an ambulance driver in a metropolitan city

Need: Needs optimized route navigation with live traffic and emergency corridor data

Insight: He regularly loses time stuck in traffic without priority routing or hospital readiness information

POV Statement:

Raju needs a smart navigation system that integrates live traffic, emergency signal preemption, and hospital availability so that he can respond faster and deliver patients in the best condition.

Dr. Meena — Emergency Room Doctor

User: Dr. Meena, an ER physician at a government hospital

Need: Needs advance notification of incoming patients with medical condition data

Insight: She frequently receives patients without preparation time, reducing the quality of immediate care

POV Statement:

Dr. Meena needs a real-time ambulance-to-hospital data link that provides patient condition updates in advance so that her team can prepare life-saving interventions before the patient arrives.

2.2 How Might We (HMW) Questions

After formulating the POV statements, we transform the core problems into open-ended, opportunity-driven questions to guide the ideation process.

HMW Questions Based on Patient and Family Needs:

6. How might we provide families with real-time ambulance tracking and ETA updates?

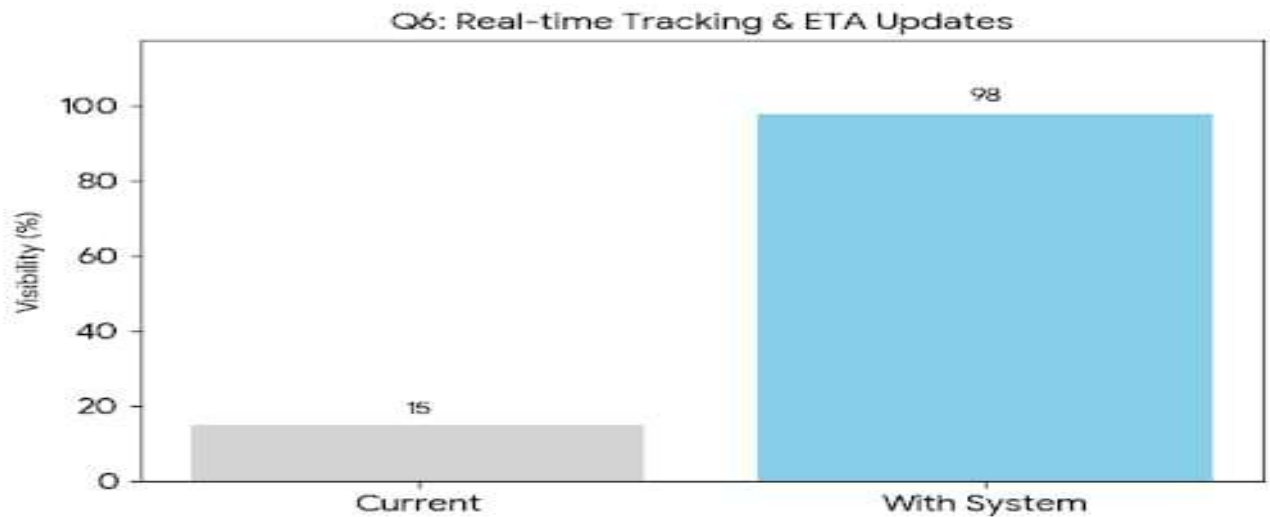


Figure Caption: Real-time Visibility Improvement. This chart illustrates the shift from a "blind" wait to 98% visibility via live GPS tracking and ETA updates for families. (Source: Generated using Type Form).

7. How might we eliminate the information void that causes panic during emergency waits?

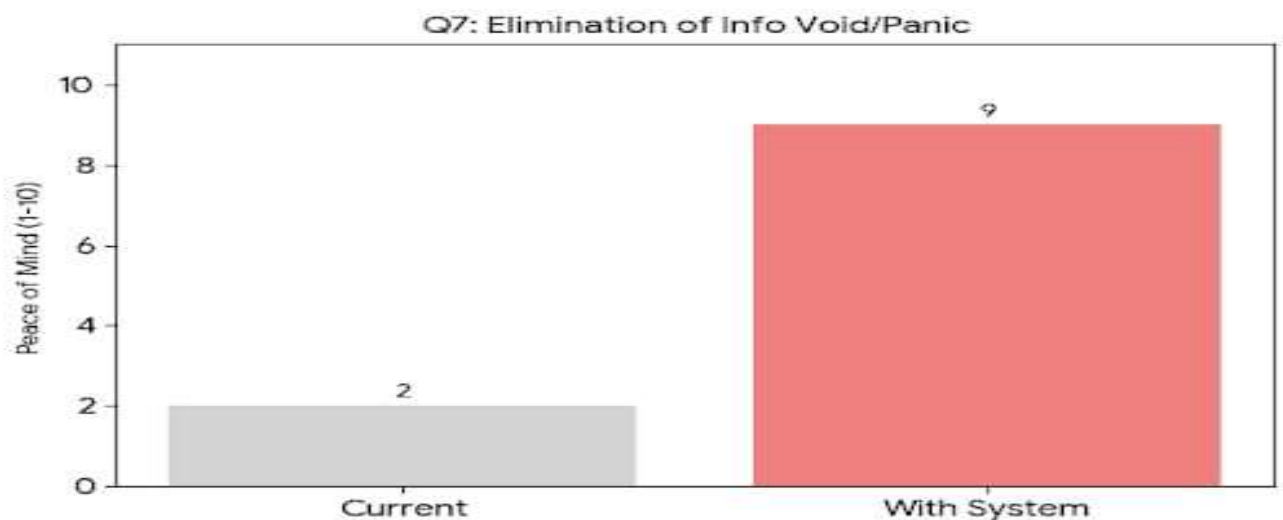


Figure Caption: Impact on Psychological Stress. A comparison showing how eliminating the "information void" increases family peace of mind from a critical low (2/10) to a stable high (9/10). (Source: Generated using Type Form).

8. How might we enable one-tap SOS with automatic location sharing for all patients?

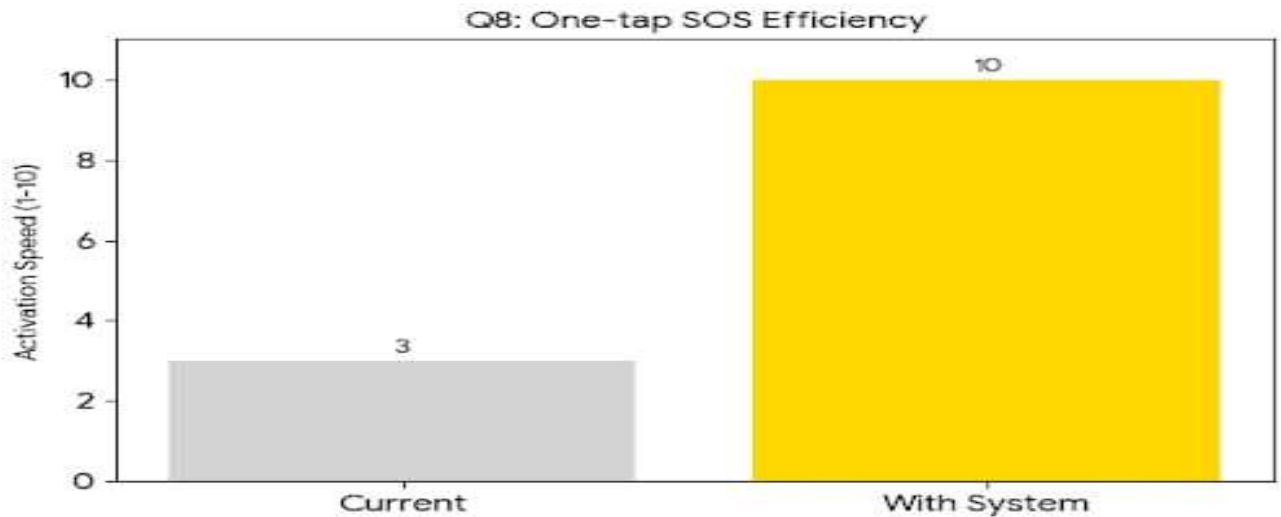


Figure Caption: SOS Activation Efficiency. Demonstrates the near-instantaneous speed of one-tap SOS and automated location sharing compared to traditional manual reporting. (Source: Generated using Type Form).

9. How might we ensure the ambulance always reaches the correct address without miscommunication?

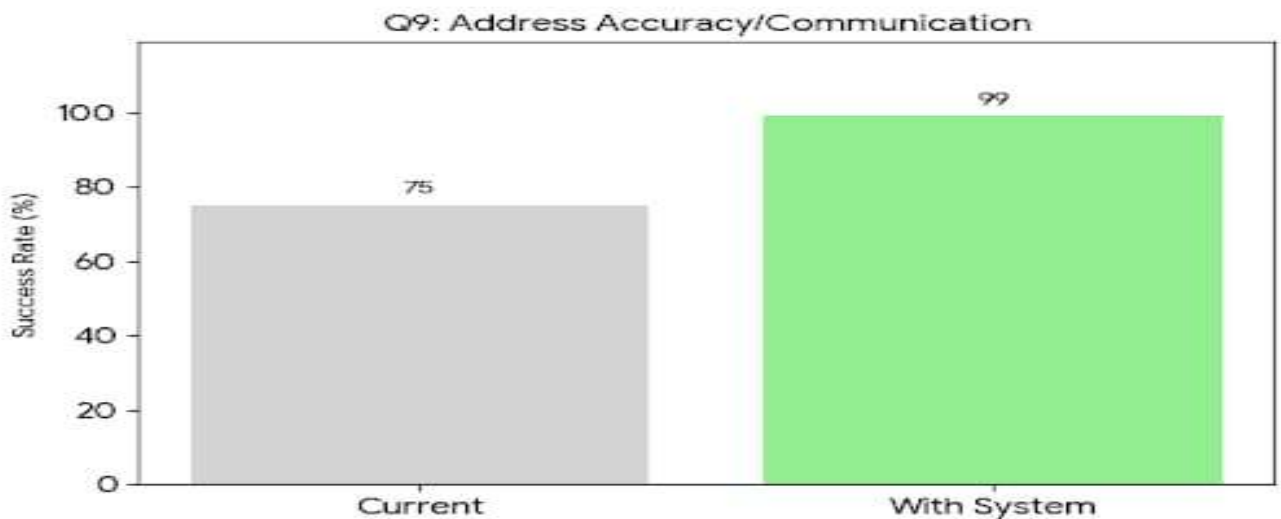


Figure Caption: Address Accuracy & Success Rate. Highlights the reduction in miscommunication, pushing address arrival success rates from 75% to near-total accuracy (99%) (Source: Generated using Type Form).

10. How might we send proactive status notifications to keep families informed throughout?

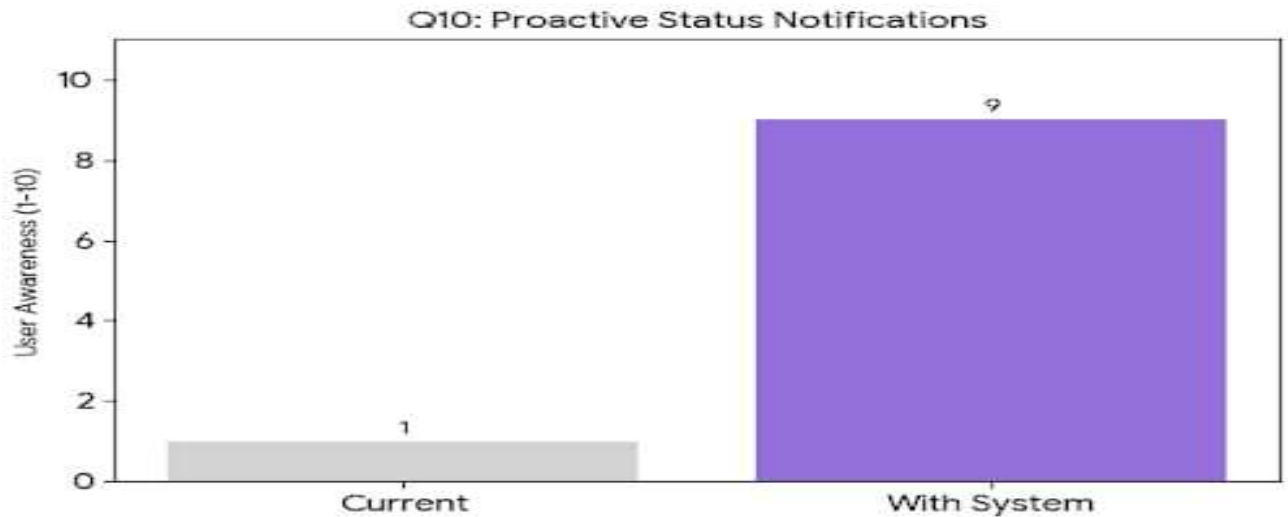


Figure Caption: Proactive Notification Engagement. Shows the jump in family awareness levels through automated status alerts versus reactive phone-based updates(**Source:**Generated using Type Form).

HMW Questions Based on Dispatcher and Driver Insights:

11. How might we automate ambulance assignment using proximity and real-time availability data?

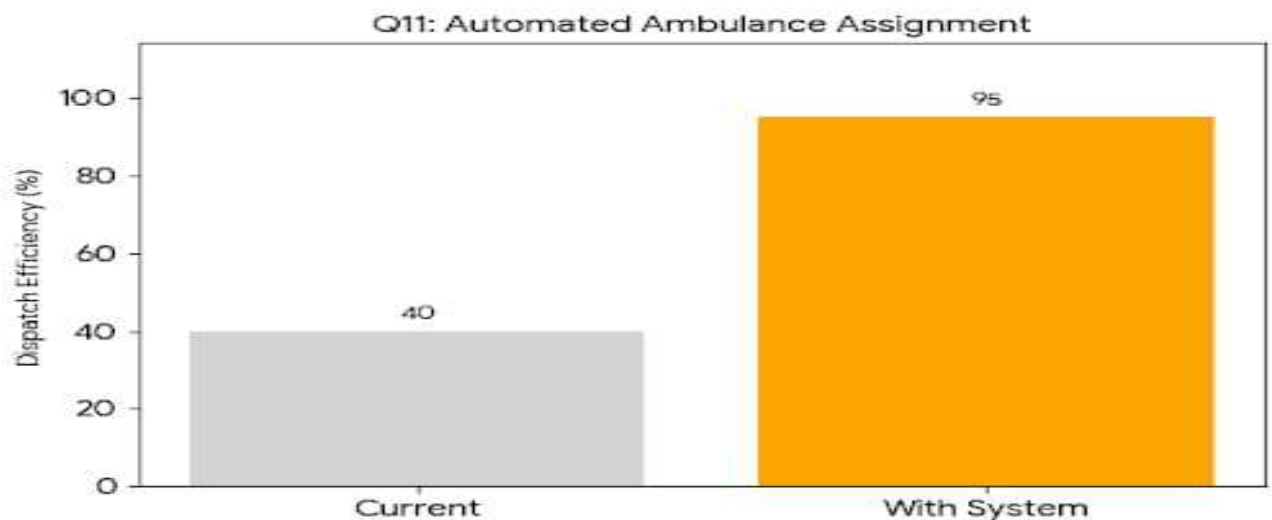


Figure Caption: Automated Dispatch Efficiency. Quantifies the improvement in assignment speed and accuracy when using proximity-based data over manual dispatching.(**Source:**Generated using Type Form).

12. How might we provide drivers with dynamic emergency routing that bypasses traffic?

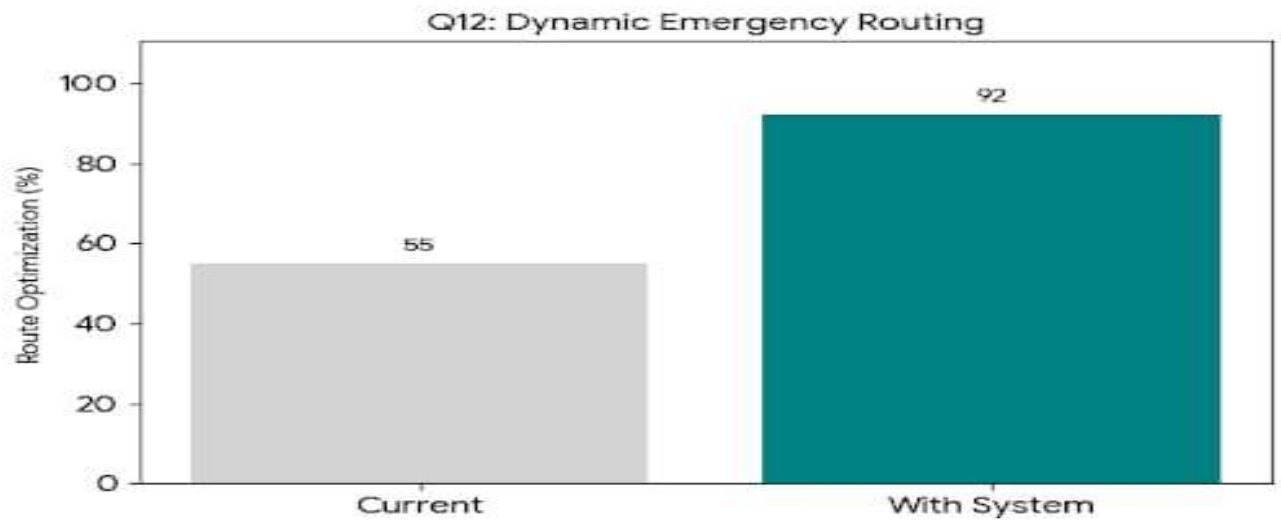


Figure Caption: Route Optimization Gains. Illustrates the effectiveness of dynamic emergency routing in bypassing traffic compared to standard navigation methods. (Source: Generated using Type Form).

13. How might we give dispatchers a live fleet overview to manage multiple emergencies at once?

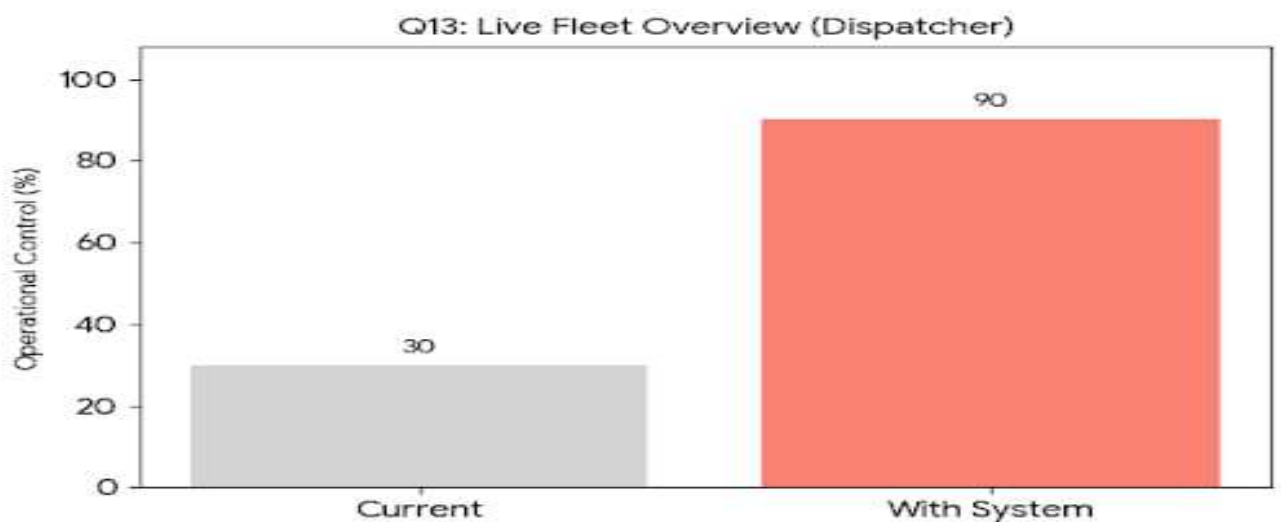


Figure Caption: Dispatcher Fleet Control. A measure of situational awareness, showing a 60% increase in a dispatcher's ability to manage multiple emergencies simultaneously. (Source: Generated using Type Form).

14. How might we enable in-ambulance communication of patient vitals to the receiving hospital?

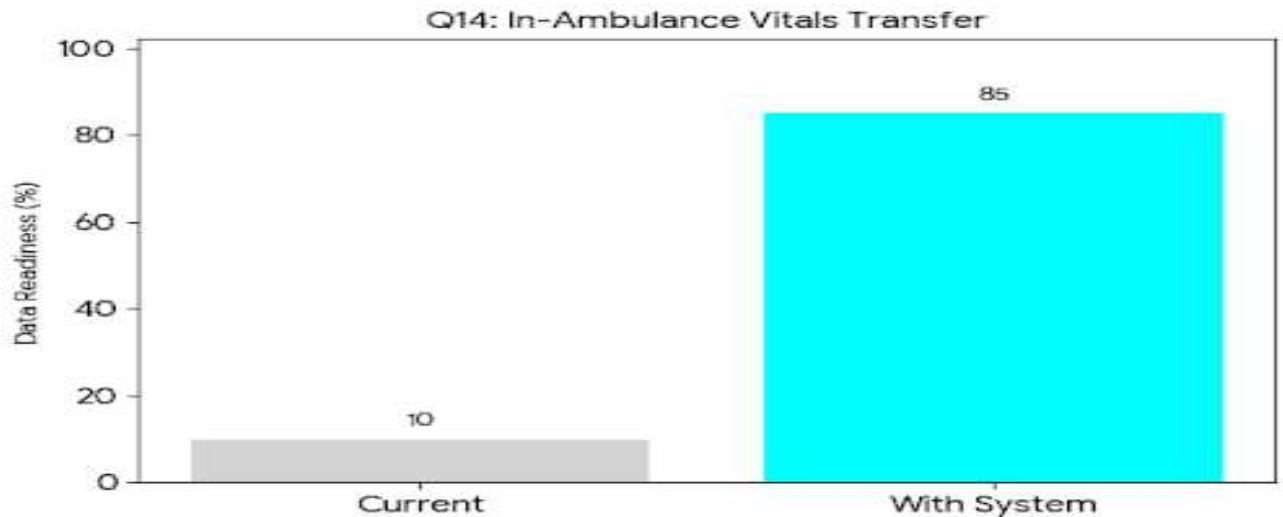


Figure Caption: Pre-Arrival Data Readiness. Contrasts the current "handoff-only" vitals model with the proposed system's 85% pre-arrival data synchronization(**Source:**Generated using Type Form).

HMW Questions Based on Hospital Staff Expectations:

15. How might we give hospitals real-time incoming patient data to prepare in advance?

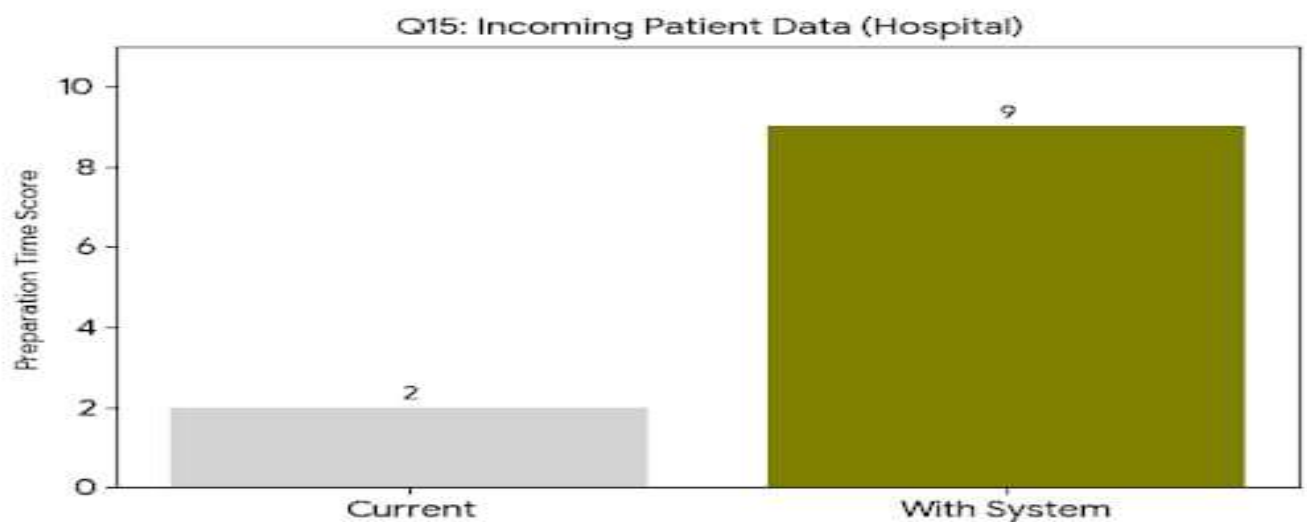


Figure Caption: Hospital Preparation Score. Tracks the shift from reactive to proactive triage, significantly improving the "preparation time" score for medical teams. (Source: Generated using Type Form).

16. How might we integrate bed availability and specialist schedules into the dispatch system?

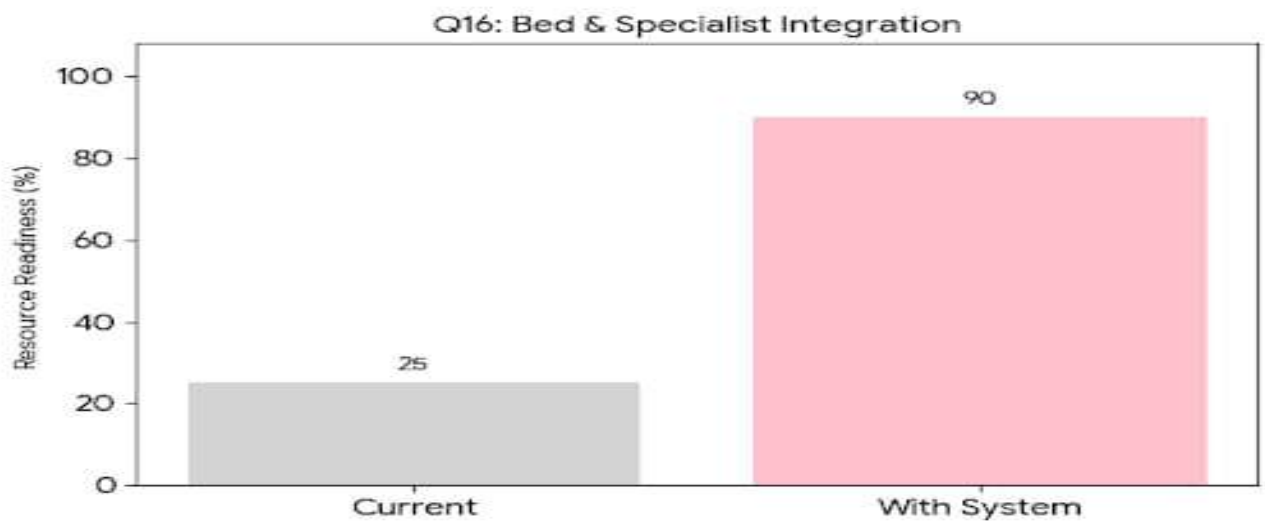


Figure Caption: Resource Allocation Accuracy. Displays the increase in hospital readiness when bed availability and specialist schedules are integrated directly into the dispatch flow. (Source: Generated using Type Form).

2.3 Journey Maps / User Story Mapping

CUSTOMER JOURNEY MAP

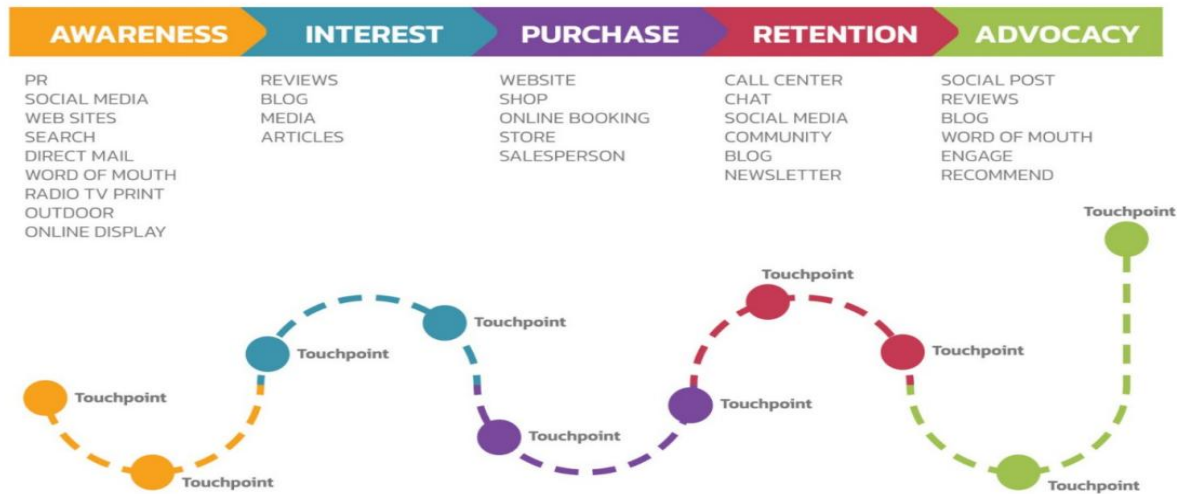


Figure 2.3: Mapping the Patient Family's Path from Panic and Helplessness to Reassurance and Safety (Source: Generated using Gemini AI)

2.4 Problem Trees

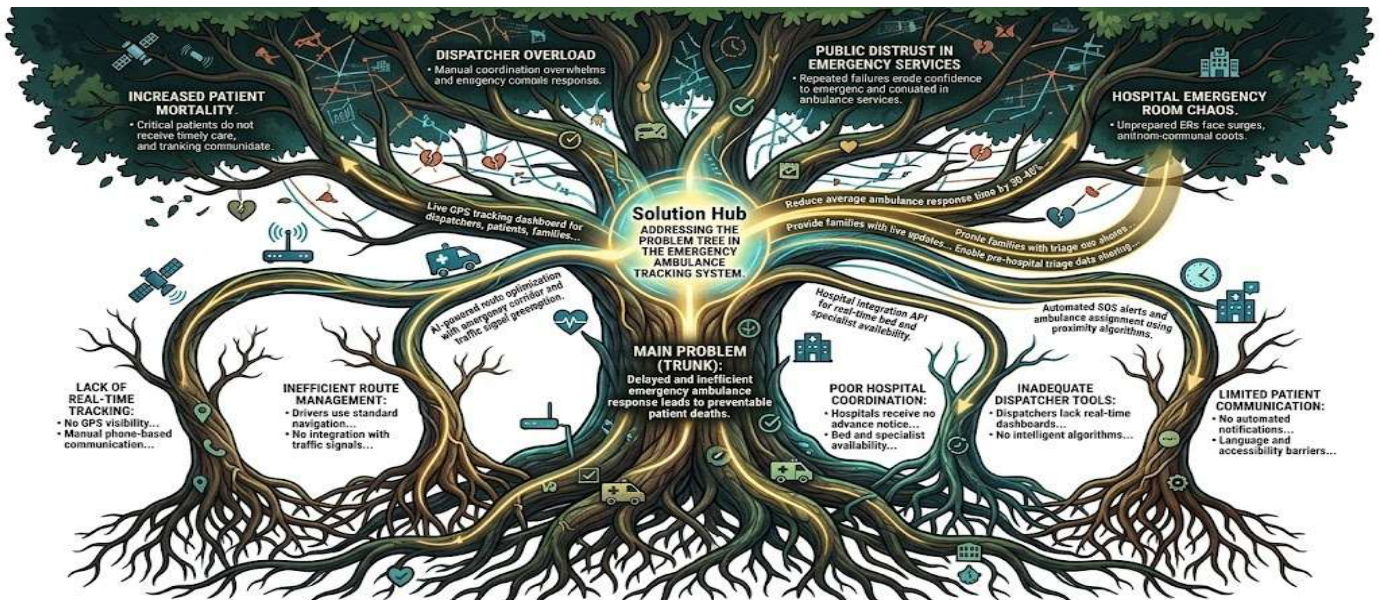


Figure 2.4: Visualizing the User Experience (UX) for Emergency Assistance Requests (Source: Generated using Gemini AI)

Introduction to Problem Trees:

A problem tree is a visual tool used to break down complex issues into root causes, effects, and underlying problems. In the Emergency Ambulance Tracking context, problem trees help identify why ambulance response systems fail and how a technology-driven solution can address those failures.

Main Problem (Trunk):

Delayed and inefficient emergency ambulance response leads to preventable patient deaths.

Root Causes (Roots):

Lack of Real-Time Tracking:

- No GPS visibility for patients, families, or dispatchers once an ambulance is dispatched.
- Manual phone-based communication introduces delays and errors.

Inefficient Route Management:

- Drivers use standard navigation apps not optimized for emergency corridors.
- No integration with traffic signal systems for emergency preemption.

Poor Hospital Coordination:

- Hospitals receive no advance notice of incoming patients or their condition.
- Bed and specialist availability is not visible to dispatchers or drivers.

Inadequate Dispatcher Tools:

- Dispatchers lack real-time fleet dashboards for multi-unit coordination.
- No intelligent algorithms for optimal ambulance assignment based on proximity and availability.

Limited Patient Communication:

- No automated notifications sent to patients or families about ambulance status.
- Language and accessibility barriers in existing emergency communication.

Effects (Branches):

Increased Patient Mortality:

- Critical patients do not receive timely care, particularly in cardiac, trauma, or stroke cases.

Dispatcher Overload:

- Manual coordination overwhelms dispatchers, causing assignment errors.

Public Distrust in Emergency Services:

- Repeated failures erode public confidence in the 108/emergency ambulance system.

Hospital Emergency Room Chaos:

- Unprepared ERs face surges without advance patient data, reducing care quality.

Addressing the Problem Tree in the Emergency Ambulance Tracking System:

Addressing Root Causes:

- Live GPS tracking dashboard for dispatchers, patients, and families via a mobile app.
- AI-powered route optimization with emergency corridor and traffic signal preemption.
- Hospital integration API for real-time bed and specialist availability.
- Automated SOS alerts and ambulance assignment using proximity algorithms.

Addressing the Effects:

- Reduce average ambulance response time by 30–40% through intelligent routing.
- Provide families with live updates every 60 seconds to reduce panic.
- Enable pre-hospital triage data sharing to prepare ER teams before patient arrival.

3. Ideate

Mind Mapping — Structuring Ideas to Understand the Whole System

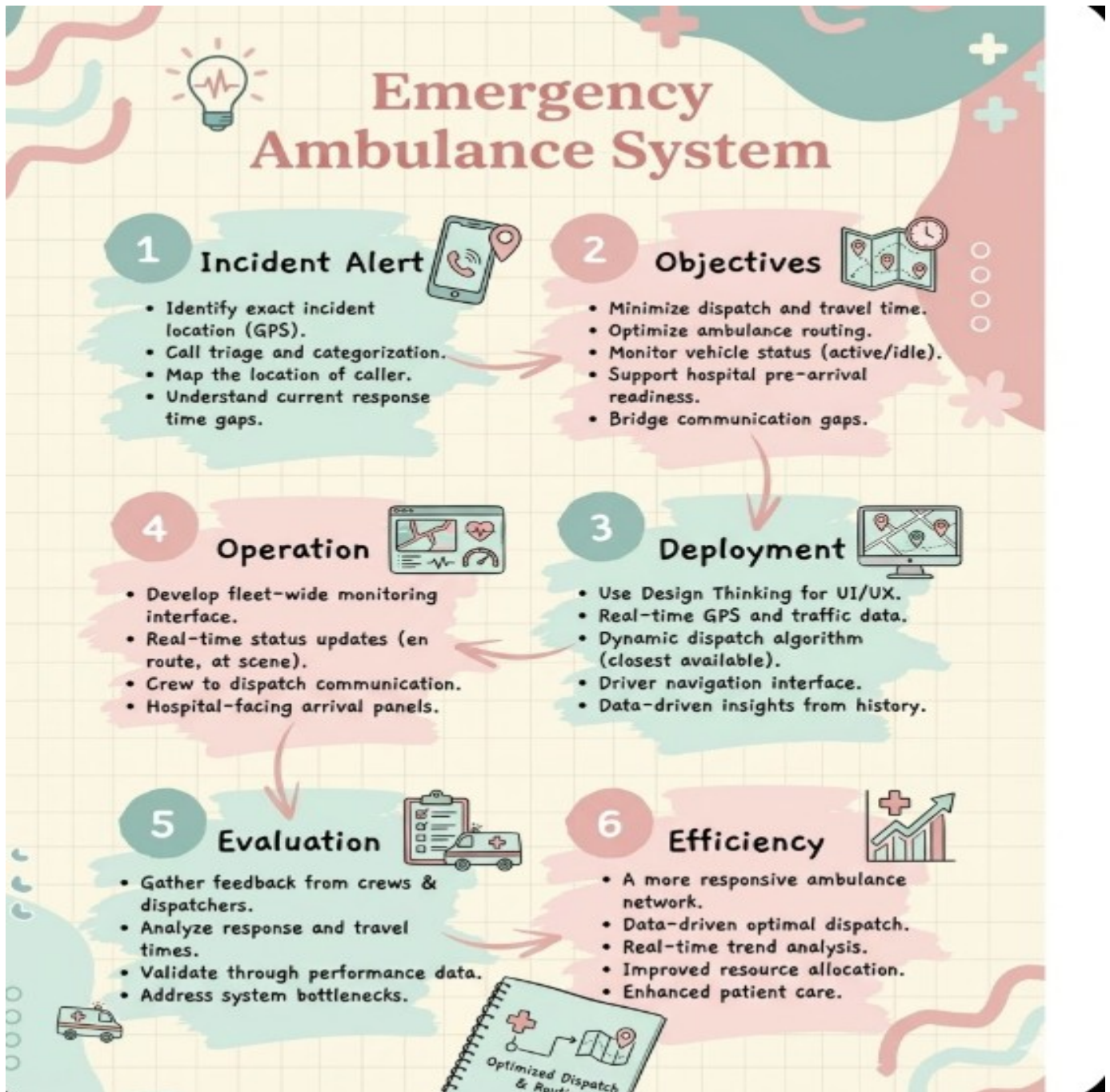


Figure Caption: Real-time Ambulance Tracking and Family Notification Interface. (Source: Generated using Gemini AI)

Storyboarding — Creating a Visual Sequence of the User Journey

Introduction to Storyboarding:

Storyboarding visualizes the sequence of events in the Emergency Ambulance Tracking System, illustrating how users interact with the application during a real medical emergency from the moment of SOS to hospital arrival.

User Scenario: A Cardiac Emergency with "Meera"

Let us consider Meera, a 32-year-old whose elderly mother suddenly collapses at home with suspected cardiac arrest.

Scene 1: SOS Activation

- Visual: Meera opens the Emergency Ambulance app in panic and presses the large red SOS button.
- Action: The app auto-detects her GPS location and sends an emergency dispatch request.
- Outcome: Nearest available ambulance is assigned within 30 seconds. Meera sees a confirmation with the ambulance's details and driver photo.

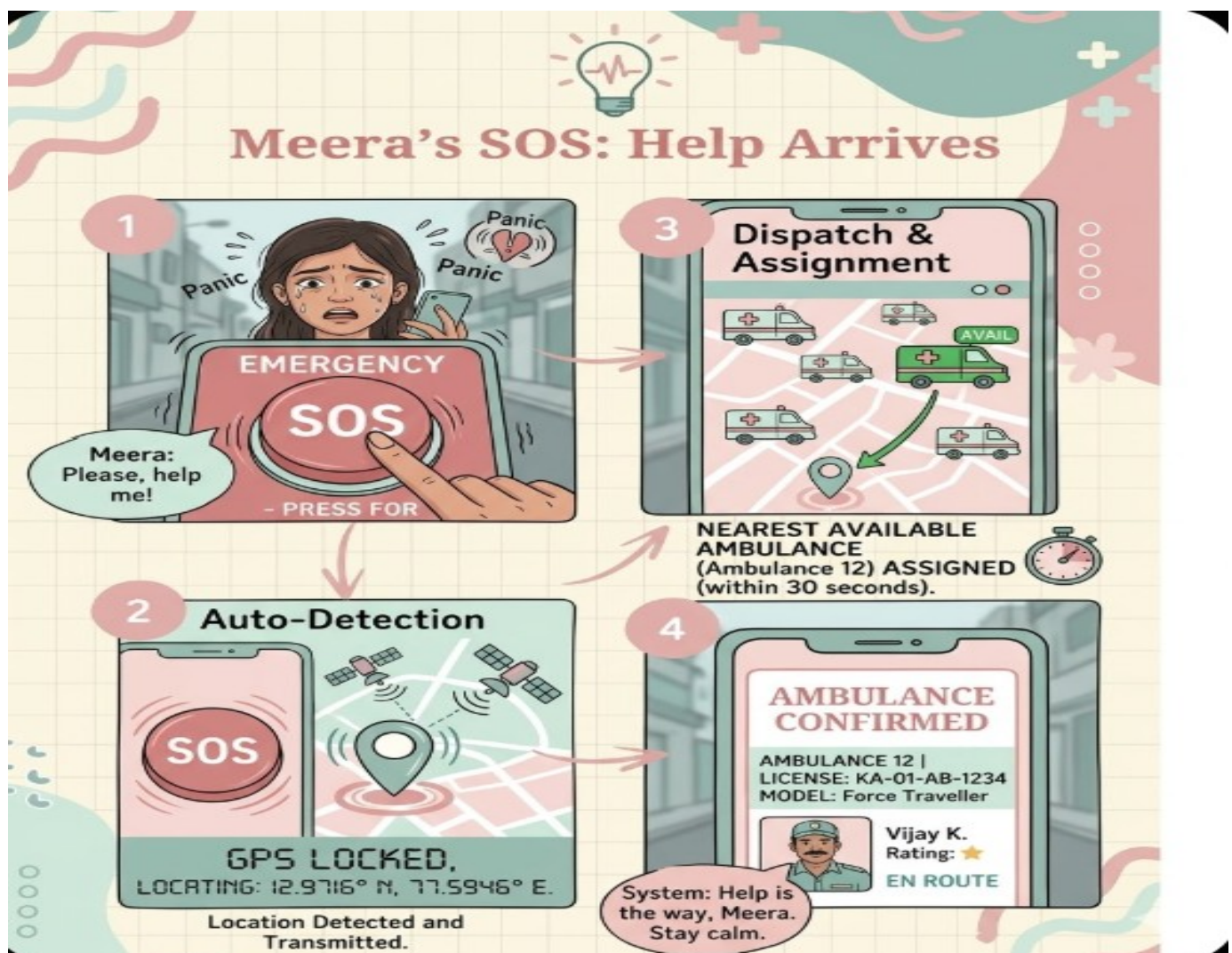


Figure Caption: *Dispatch Control Panel with Unified Dashboard and Situational Monitoring. (Source: Generated using Gemini AI)*

Scene 2: Live Ambulance Tracking

- Visual: A map appears on Meera's screen showing the ambulance's real-time location moving towards her.
- Action: The app displays ETA, ambulance ID, and driver contact. Meera shares the tracking link with other family members.

Outcome: Anxiety is significantly reduced as Meera can see the ambulance approaching in real-time



Figure Caption: Optimized Route Navigation and Traffic Bypass for Reduced Response Time. (Source: Generated using Gemini AI)

Scene 3: Dispatch Control Panel

- Visual: The dispatcher's dashboard shows all active ambulances, emergencies, and hospital bed availability in one view.

- Action: Dispatcher reroutes a second ambulance to cover another incoming emergency.

Outcome: Efficient multi-unit management without phone calls or manual coordination.

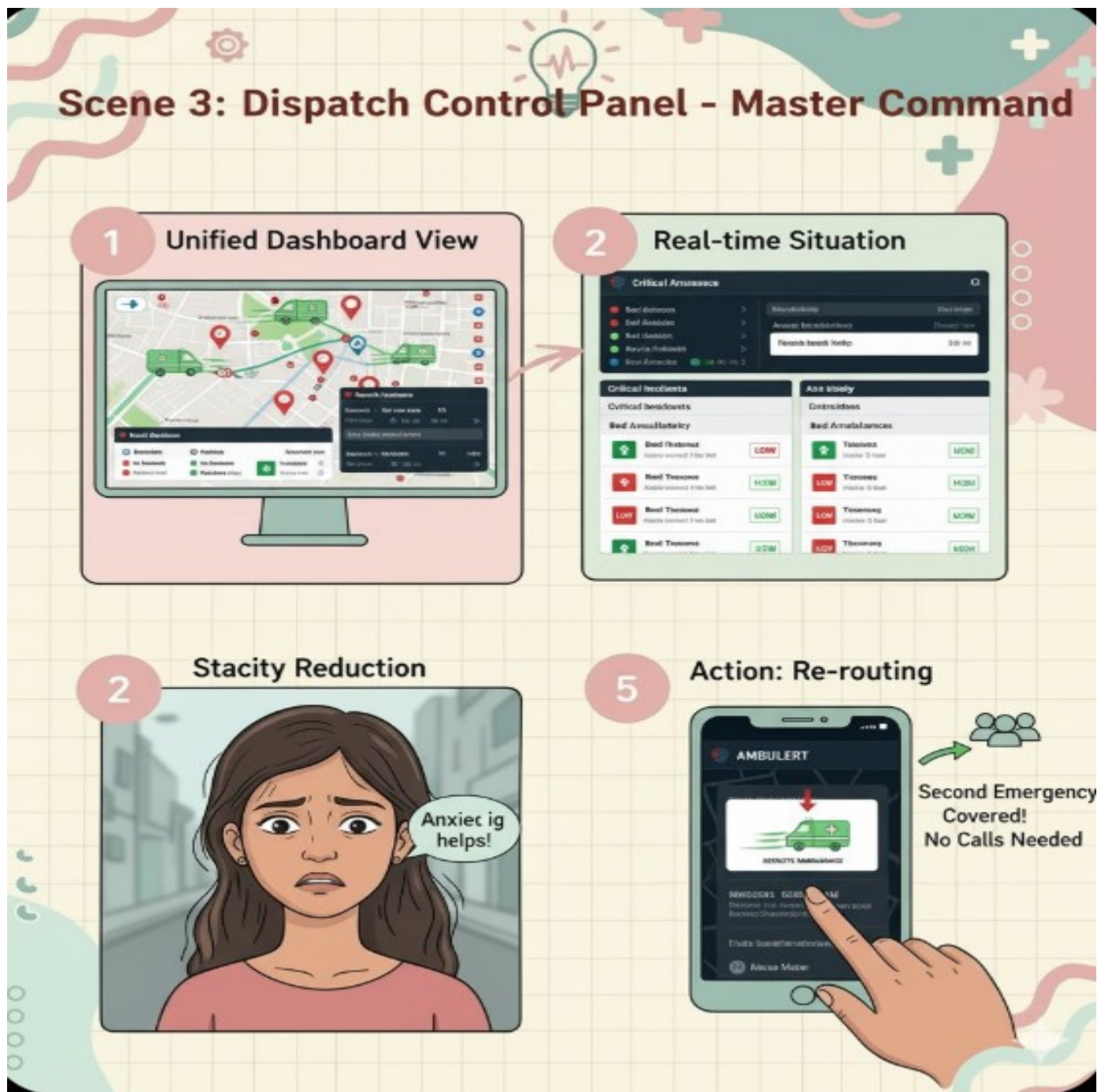


Figure Caption: Hospital Readiness Phase: Automated Alerts and Data Transfer to ER Teams. (Source: Generated using Gemini AI)

Scene 4: Driver Navigation

- Visual: The driver sees an optimized emergency route on the in-vehicle screen, with traffic signal preemption activated.
- Action: Driver follows the route, bypassing a traffic jam via an emergency corridor.
- Outcome: Response time is reduced by 12 minutes compared to standard GPS navigation.

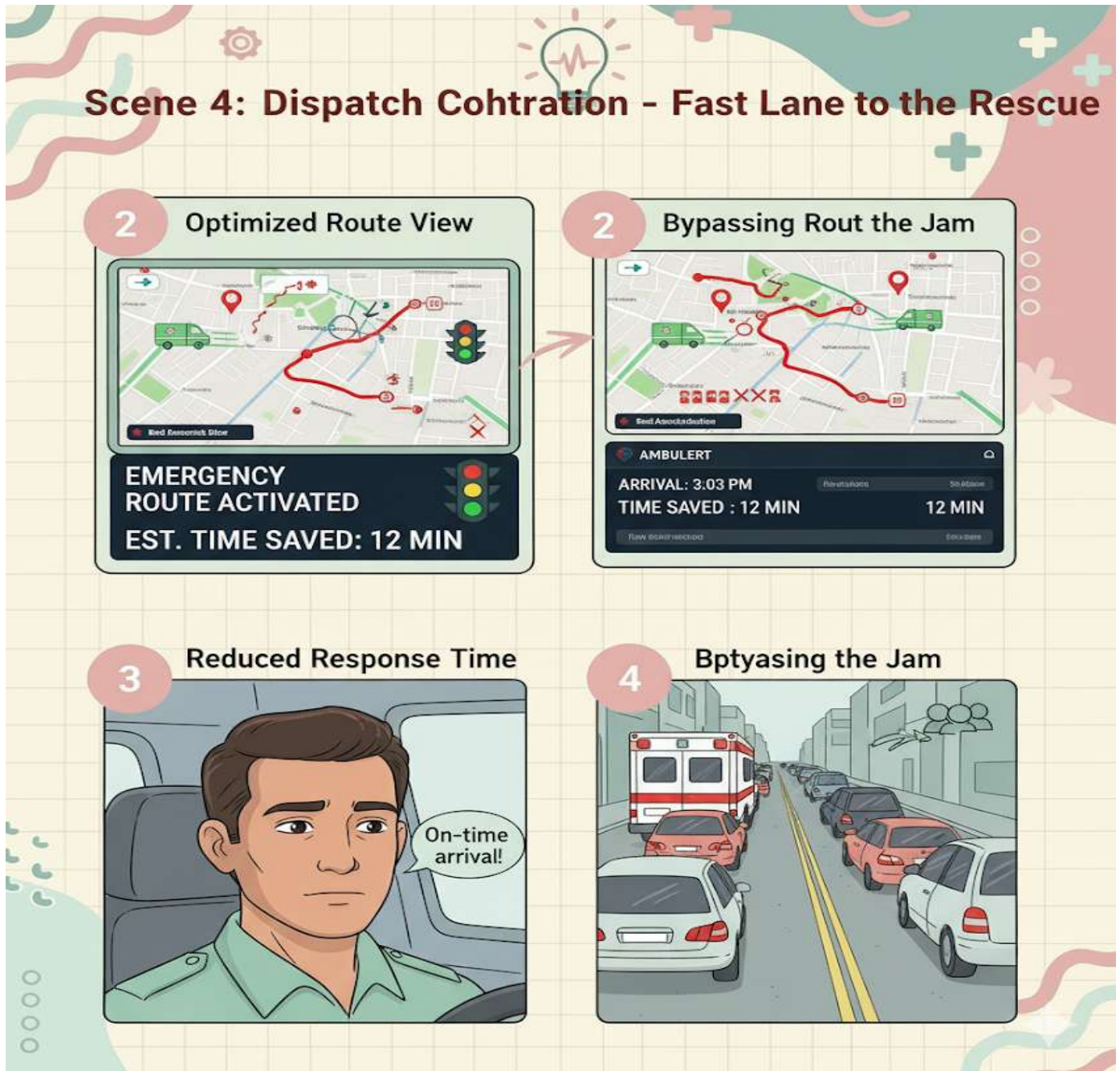


Figure Caption: Family Communication Interface Featuring Real-time Status Updates and Timely Handoff. (Source: Generated using Gemini AI)

Scene 5: Hospital Notification

- Visual: The receiving hospital's ER dashboard receives an automatic alert with patient details, ETA, and preliminary vitals.
- Action: ER team prepares a trauma bay and calls the relevant specialist.
- Outcome: Hospital is fully prepared 8 minutes before patient arrival, reducing time-to-treatment.

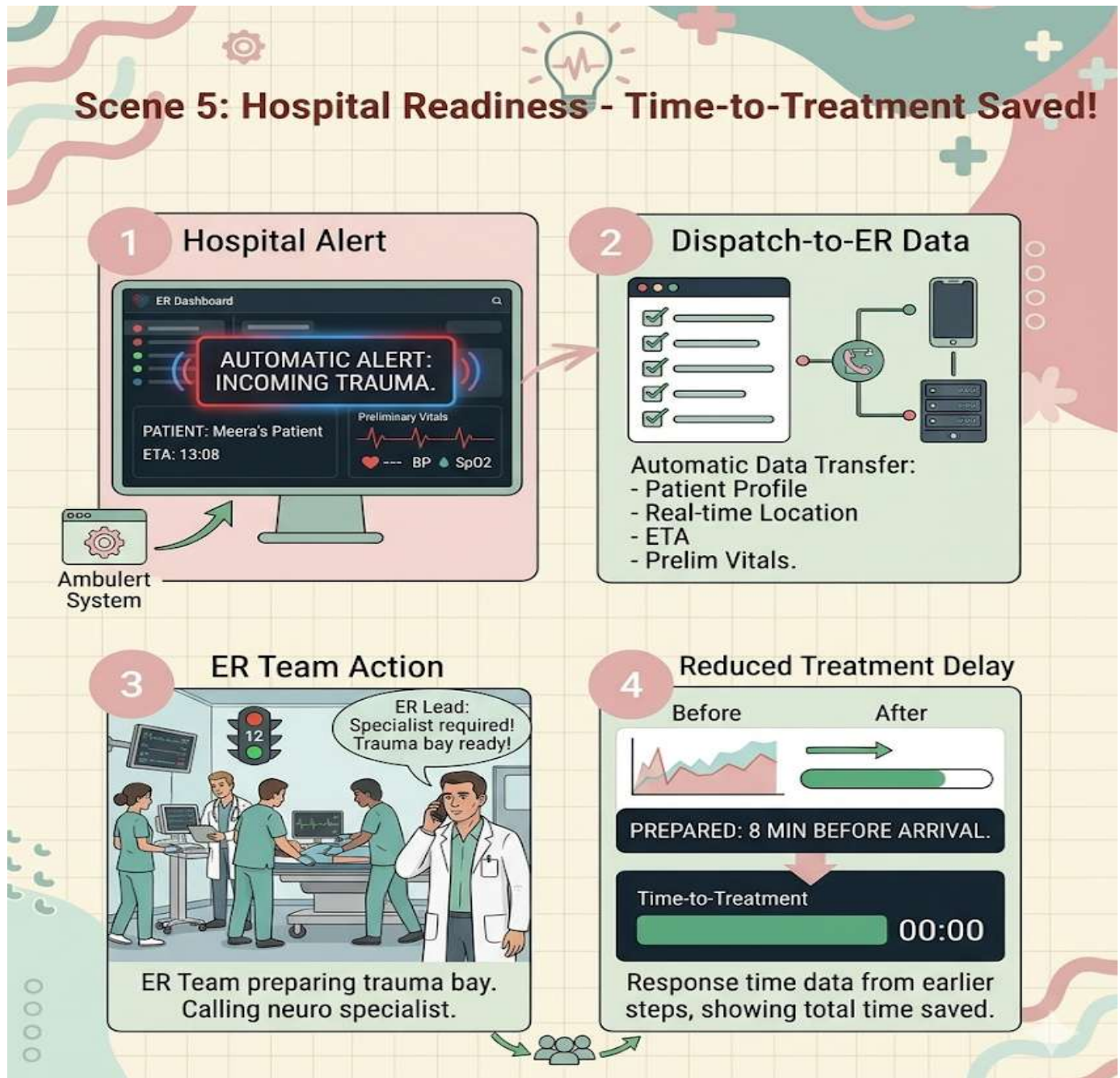


Figure Caption: Post-Trip Reporting and the Continuous Improvement Cycle for Emergency Services. (Source: Generated using Gemini AI)

Scene 6: Status Updates to Family

- Visual: Meera receives push notifications — 'Ambulance 5 minutes away', 'Patient loaded', 'En route to City General Hospital'.
- Action: Family members follow the ambulance live on the shared tracking link.
- Outcome: Complete transparency eliminates anxiety and ensures family reaches the hospital in time.

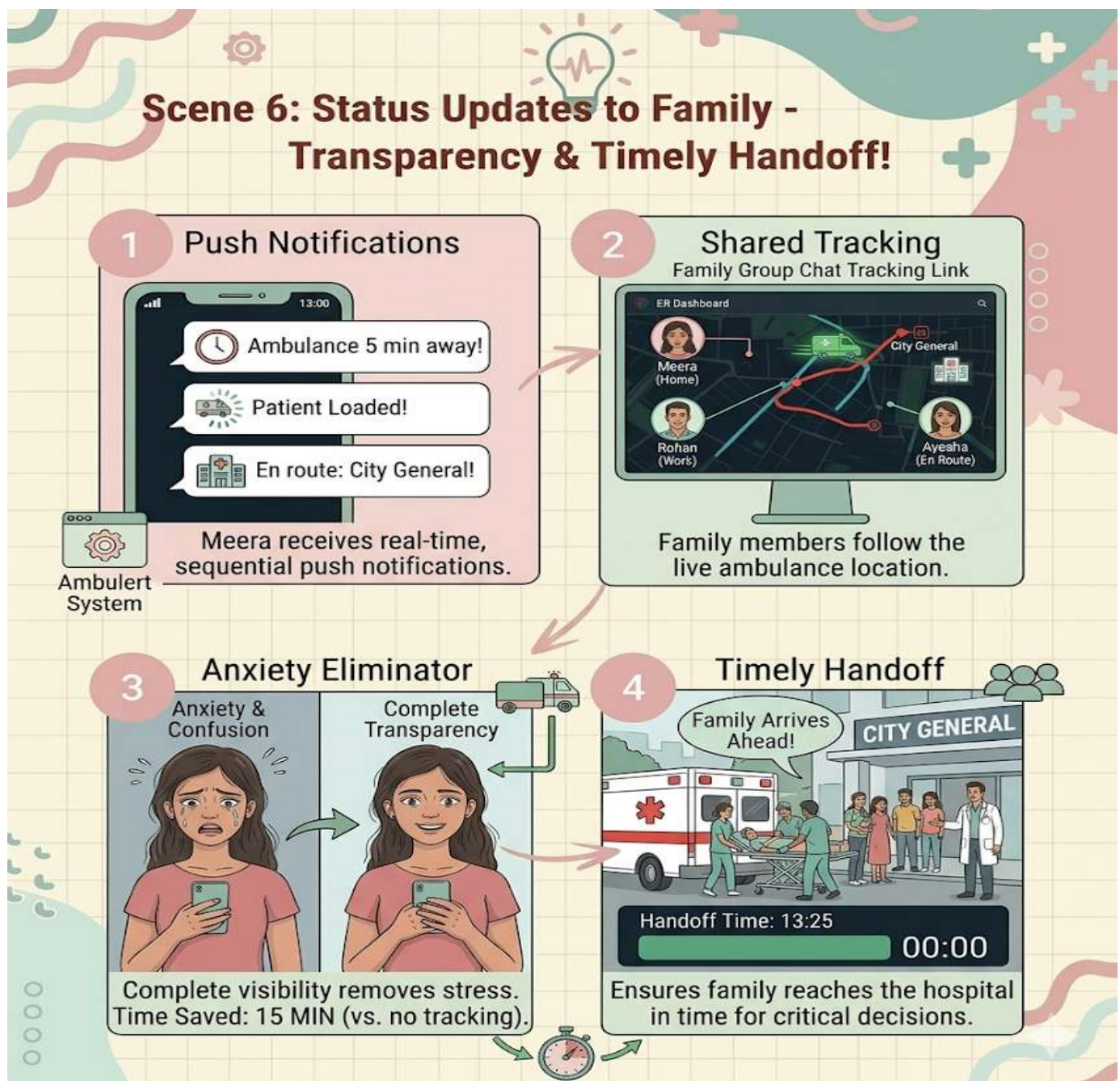


Figure Caption:Emergency Ambulance Tracking System workflow.(Source: Generated using Gemini AI)

Scene 7: Post-Trip Report

- Visual: After the emergency, the app generates a full trip report: response time, route taken, hospital, and care summary.
- Action: Family provides feedback. Data feeds into system analytics for continuous improvement.
- Outcome: Operational insights help improve future emergency responses in the city.

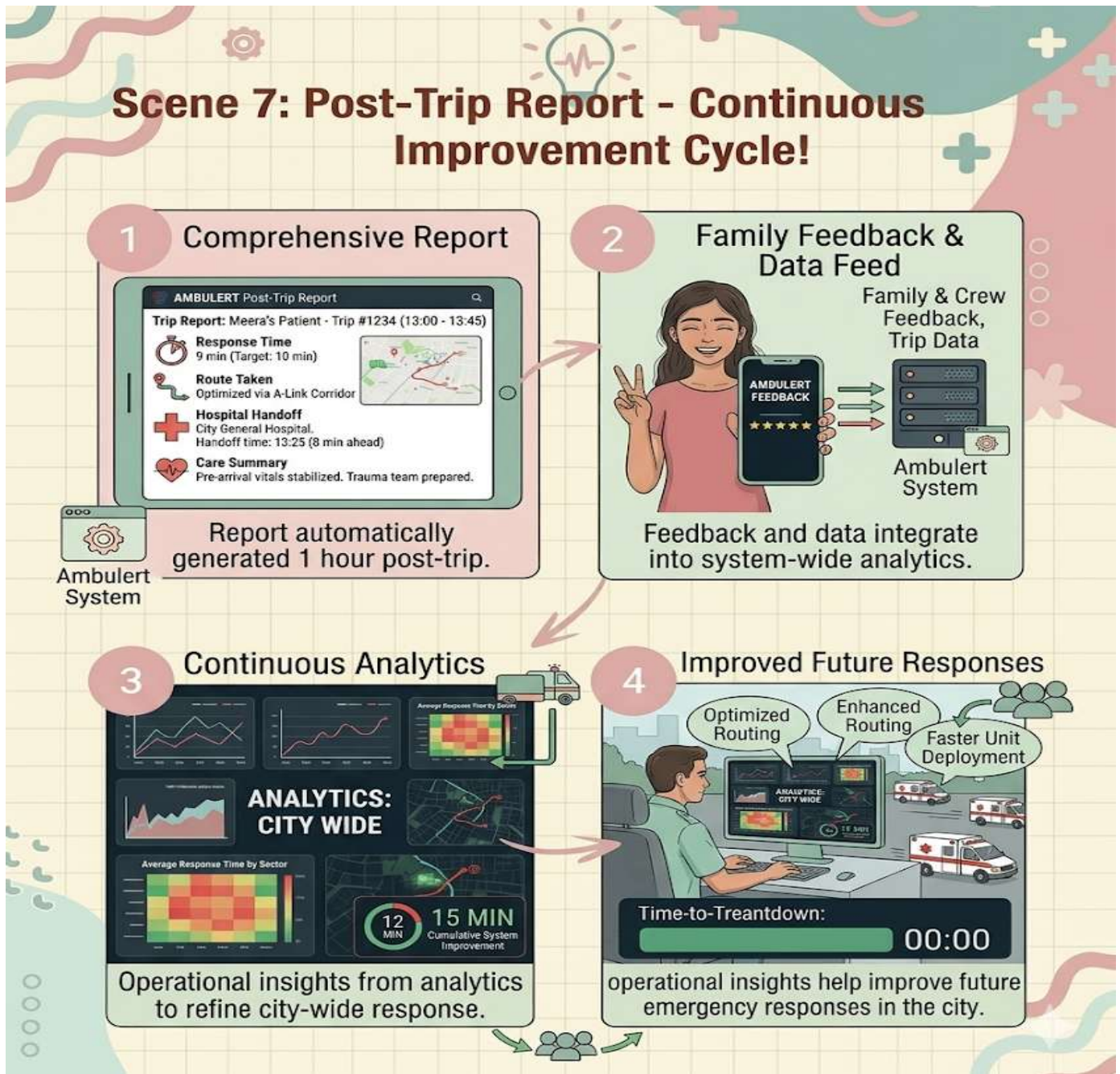


Figure Caption: Post_Trip Analysis & Improvement this stage highlights the continuous improvement cycle. (Source: Generated using Gemini AI)

4. Prototype

Click-through Prototypes

Introduction to Click-through Prototypes:

Click-through prototypes are interactive digital mockups that allow users to simulate the navigation and user experience of the Emergency Ambulance Tracking System. These prototypes enable designers, dispatchers, and testers to experience the flow and structure without full development.

In the case of the Emergency Ambulance Tracking System, the Click-through Prototype serves as an early version of the platform where users can interact with SOS activation, ambulance tracking maps, dispatcher dashboards, and hospital readiness screens.

Benefits of Click-through Prototypes:

Interactive Emergency Flow Simulation:

- Demonstrate how users navigate from SOS to live tracking, hospital notification, and post-trip feedback.

Realistic Emergency Experience Testing:

- Simulate the panic of an emergency and test whether the interface is usable under stress.

Cost and Time Efficiency:

- Build platform layout and interactivity in Figma or InVision before investing in full development.

Usability Testing Before Deployment:

- Test SOS flow, tracking map, and dispatcher assignment before real-world deployment.

Stakeholder Feedback and Rapid Iteration:

- Share with dispatchers, drivers, and hospital staff to gather targeted feedback quickly.

Tools to Use:

- Figma — for interactive prototype screens and clickable flows
- InVision — for multi-user prototype sharing with comments
- Adobe XD — for advanced animation and micro-interaction prototyping

Logo for the Prototype — AMBULERT (Emergency Ambulance Tracking System):



Figure Caption: Official Brand Identity and Logo for "AMBULERT."(Source: Generated using Gemini AI)

Wireframing:

- Key screens: SOS activation, Live Map Tracking, Driver Navigation, Dispatcher Dashboard, Hospital Readiness Panel, Notification Center, Trip Report.
- Use simple layouts to outline functionality before high-fidelity designs.



Figure Caption: Initial Wireframes and Interactive Flow Simulation (Source: Generated using Gemini AI)

Design High-Fidelity Mockups:

- Apply emergency-themed UI: red SOS button, live green ambulance icon, clear map overlays.
- Use high contrast, large fonts, and minimal input for use under high-stress conditions.

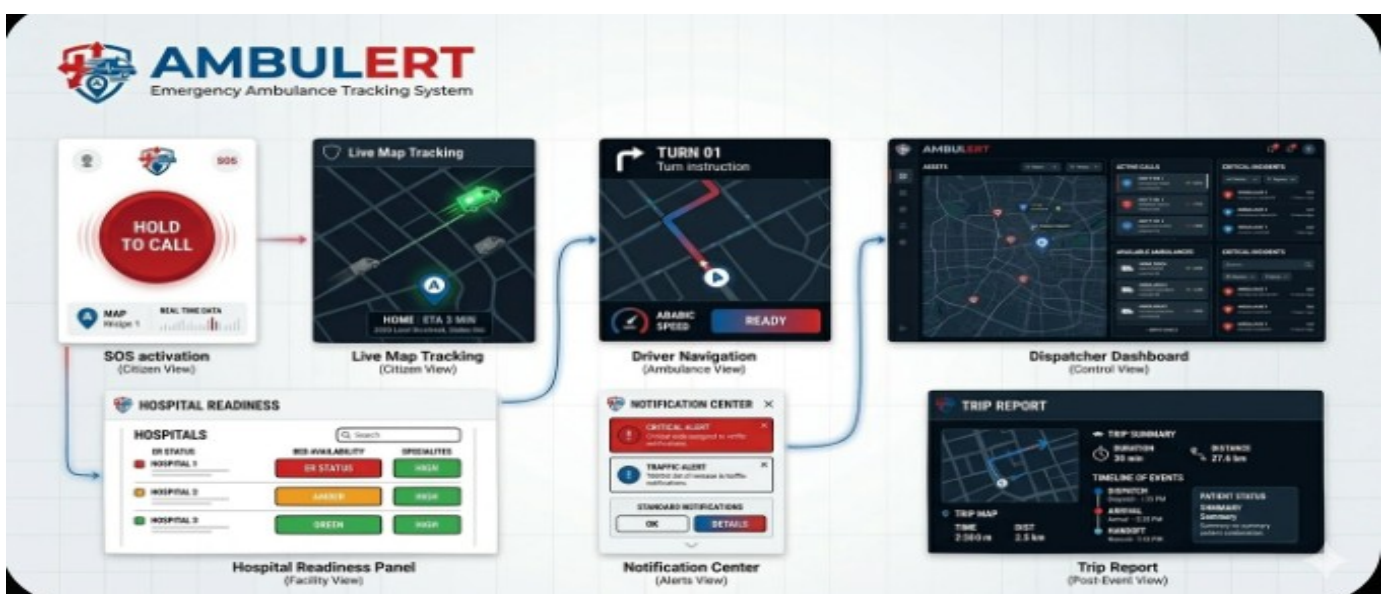


Figure Caption: High-Fidelity UI Design Featuring High-Contrast Visuals for Stressful Environments. (Source: Generated using Gemini AI)

Interactive Links and Buttons:

- Link SOS button to map screen, dispatcher assignment to driver navigation, notifications to family tracking view.
- Simulate ambulance movement on map using animation frames in Figma.



Figure Caption: Full-Scale Interactive Flow Simulation of the AMBULERT Ecosystem. (Source: Generated using Gemini AI)

User Testing:

- Share prototype with family members, drivers, and dispatchers for stress-condition usability testing.
- Iterate based on feedback to optimize emergency UX before system development.

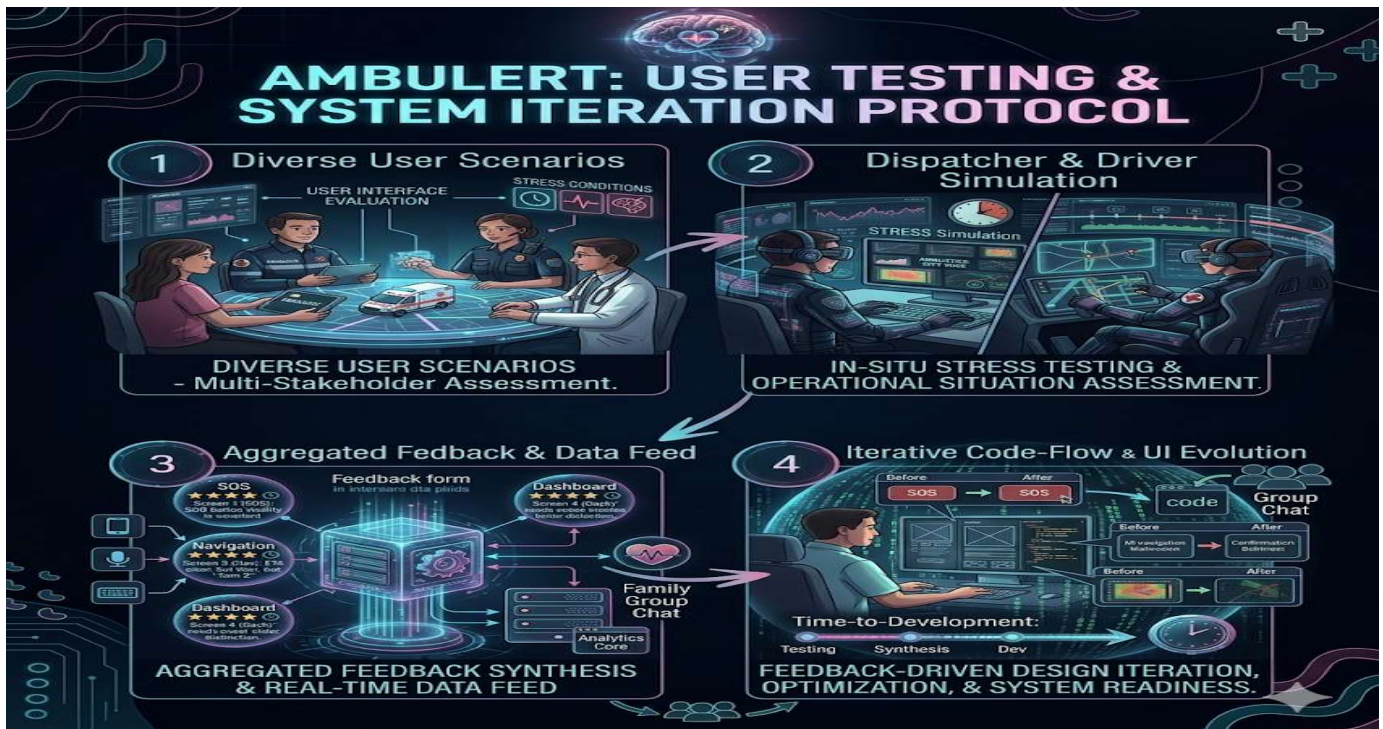


Figure Caption: Multi-Stakeholder Assessment and Feedback Integration Cycle.(Source: Generated using Gemini AI)

5. Test

Test — Usability Testing

Introduction to Usability Testing:

Usability testing is a method where real users interact with a prototype to evaluate how user-friendly and intuitive it is. In this context, usability testing allows us to observe how users — patients, families, dispatchers, and drivers — interact with the Emergency Ambulance Tracking System prototype and gather insights to improve the user experience under high-stress emergency conditions.

Why Choose Usability Testing?

- **Identify User Pain Points:** Understand where users struggle, especially during high-stress emergency scenarios.
- **Evaluate Ease of Navigation:** Test whether users can activate SOS, track ambulance, and receive notifications intuitively.

- Test Critical Functionality: Verify SOS button, live map, notifications, dispatcher assignment, and hospital alerts work properly.
- Improve Emergency UX: Ensure users can complete life-critical actions within seconds, not minutes.

Steps for Conducting Usability Testing:

Define Testing Goals:

- Can a panicked user activate SOS within 5 seconds of opening the app?
- Can a dispatcher assign the nearest ambulance within 30 seconds using the dashboard?
- Can hospital staff view incoming patient data with at least 5 minutes advance notice?

Select Participants:

- 5 patient family members with no prior tech training
- 3 ambulance dispatchers from city emergency centers
- 2 ambulance drivers
- 2 ER nurses to test hospital notification screens

Create Test Scenarios:

- "Activate an emergency SOS for a cardiac arrest patient and track the assigned ambulance."
- "As a dispatcher, assign the nearest available ambulance to an incoming emergency."
- "As a hospital ER nurse, view the incoming patient profile and prepare the appropriate bay."

Set Up the Test Environment:

- Conduct tests in simulated emergency scenarios to capture real stress-level responses.
- Use screen recording software to capture interaction data for analysis.
- Provide basic task instruction without coaching on app usage.

Conduct the Usability Test:

- Have participants complete tasks independently while observers note hesitation points.
- Use think-aloud protocol for patients and families to capture real-time confusion.
- Measure time-to-task completion for each critical action.

Collect Feedback:

- Post-test survey: Was the SOS button obvious enough? Was the map readable? Were notifications clear?
- Record quantitative data: task completion rate, error rate, and time-on-task.

Analyze Results:

- Identify most common failures: users missing SOS button, unclear map icons, notification delays.
- Prioritize fixes that directly affect emergency response time.

Make Improvements:

- Enlarge and recolor SOS button for immediate visibility.
- Add audio/vibration alerts for ambulance ETA notifications.
- Simplify dispatcher assignment to a single click with AI-auto-suggestion.

Key Areas to Focus During Usability Testing:

SOS Activation Speed:

- Can users activate the SOS in under 5 seconds, including location sharing?

Map Readability:

- Is the ambulance icon clearly visible and moving in real-time on the map?
- Are ETAs prominently displayed in large, readable format?

Notification Clarity:

- Are push notifications delivered promptly and clearly understandable?
- Do users know which hospital the patient is being taken to?

Dispatcher Dashboard Efficiency:

- Can dispatchers see all active emergencies and available ambulances simultaneously?
- Is ambulance assignment completion under 30 seconds?

Example Usability Test Scenarios:

Scenario 1: SOS Activation

- Task: Open the app and activate an SOS for a family member in cardiac arrest.
- Observation: Time to first SOS tap, errors in location sharing, confirmation screen clarity.
- Success Metric: SOS activated with location confirmed within 10 seconds.

Scenario 2: Live Ambulance Tracking

- Task: Track the dispatched ambulance on the map and identify the ETA.
- Observation: Map interaction ease, ambulance icon clarity, ETA readability.
- Success Metric: User identifies ETA within 15 seconds without assistance.

Scenario 3: Hospital Staff Notification

- Task: As ER staff, view the incoming patient profile and select the appropriate preparation protocol.
- Observation: Speed of accessing patient data, clarity of condition summary, action buttons.
- Success Metric: Staff selects preparation protocol within 60 seconds.

Prototype Access Note

To experience the Emergency Ambulance Tracking System prototype, click the link below:

<https://ambulert-demo.netlify.app/>

This interactive prototype allows you to explore:

- SOS activation and GPS location sharing
- Live ambulance tracking map with real-time ETA
- Dispatcher fleet management dashboard
- Hospital readiness and patient notification screens

Kindly visit the link and provide feedback for further improvement!

Reflections On The Phases Of Design Thinking Process

Through engaging in the Design Thinking process to create and develop the Emergency Ambulance Tracking System, I gained profound insights into how structured design methodologies can transform life-critical ideas into actionable, empathetic solutions. Here is what I learned from each phase:

Empathy Phase:

- **Learning:** The Empathy Phase was crucial in understanding the raw fear and helplessness experienced by patients and families during emergencies. It taught me that designing a life-critical system requires deeply empathetic research — understanding trauma, panic, and time pressure.
- **Key Takeaways:** Interacting directly with families who had experienced emergency delays, and with frontline drivers and dispatchers, revealed that the most important feature is not technology sophistication but transparency — knowing where help is.

Define Phase:

- **Learning:** Framing the problem through POV statements helped us see that the real issue is not just 'slow ambulances' but a complete breakdown in communication, routing intelligence, and inter-system integration.
- **Key Takeaways:** HMW questions unlocked creative thinking. 'How might we give hospitals a 5-minute preparation window?' directly inspired the pre-hospital triage notification feature.

Ideate Phase:

- **Learning:** The Ideate Phase reinforced that creativity is essential even in technical emergency systems. Storyboarding Meera's scenario helped the team visualize how every second of delay translates to human cost.
- **Key Takeaways:** Combining mind mapping with storyboarding allowed us to discover system integration opportunities we had not initially considered, such as linking signal preemption to route optimization.

Prototype Phase:

- **Learning:** Click-through prototyping revealed that emergency UX requires different design principles — fewer taps, larger buttons, audio alerts, and offline functionality for poor network zones.
- **Key Takeaways:** Failing early with a low-fidelity prototype saved significant development time. Users identified critical flaws in the SOS flow that would have been expensive to fix post-development.

Test Phase:

- **Learning:** Usability testing under simulated emergency conditions exposed that what seems intuitive in calm conditions fails under panic. The SOS button needed to be 3x larger based on testing feedback.

- **Key Takeaways:** Continuous testing and user iteration are not optional in life-critical systems — they are mandatory. Every design decision must be validated against emergency-scenario performance.

Overall Reflections:

This experience taught me the importance of adopting a deeply human-centered approach in systems where lives are at stake. The Design Thinking methodology revealed that solving the emergency ambulance response problem is not purely a technology challenge — it is a human experience design challenge. Each phase built upon the last, ensuring that the final product is grounded in real user needs, validated by frontline stakeholders, and optimized for performance under the most stressful conditions imaginable.

Going forward, I will incorporate these design thinking principles in all future engineering and technology projects, knowing that the user's experience — especially in crisis moments — must be at the center of every design decision.

Reflections On Technical & Non-Technical Issues In Ambulert Development

1. Technical Challenges:

Real-Time GPS Integration

- **Challenge:** Achieving sub-second GPS accuracy in dense urban environments with tall buildings affecting signal quality.
- **Solution:** Used multi-source GPS (satellite + cell tower triangulation) and implemented Kalman filter smoothing for accurate location prediction.
- **Learning:** Real-time location systems require redundant positioning methods for reliable emergency-grade accuracy.

Route Optimization Under Dynamic Conditions

- **Challenge:** Traffic conditions change second-by-second; static route suggestions quickly become outdated during emergencies.

- Solution: Integrated live traffic APIs (Google Maps Platform, HERE Maps) with a custom emergency corridor priority algorithm.
- Learning: Emergency routing requires a separate priority layer above standard navigation — ambulances must have algorithmically enforced route priority.

Multi-System Integration

- Challenge: Connecting ambulance GPS, dispatcher dashboard, hospital management systems, and patient mobile app into a unified real-time platform.
- Solution: Built a central API gateway with WebSocket connections for real-time bidirectional data flow across all modules.
- Learning: System integration in emergency platforms demands zero-latency communication architecture and robust fallback mechanisms.

Offline Functionality

- Challenge: Ambulances frequently pass through low-connectivity zones, especially in rural and peri-urban areas.
- Solution: Implemented offline-first architecture with local data caching and automatic sync when connectivity resumes.
- Learning: Life-critical apps must be designed for connectivity-failure scenarios from day one.

2. Non-Technical Challenges:

Stakeholder Trust and Adoption

- Challenge: Hospitals and government health bodies were initially skeptical about adopting a new dispatch technology.
- Solution: Conducted pilot deployments with measurable KPIs (response time improvement, fatality reduction rate) and shared results transparently.
- Learning: Trust in life-critical systems is earned through demonstrated results, not marketing claims.

Driver and Dispatcher Resistance

- Challenge: Experienced drivers and dispatchers resisted changing established workflows, even if those workflows were inefficient.

- Solution: Held co-design workshops where their expertise shaped the interface, making adoption feel like empowerment rather than replacement.
- Learning: Technology adoption in emergency services succeeds when frontline workers feel ownership of the solution.

Data Privacy and Patient Confidentiality

- Challenge: Real-time patient tracking raises significant privacy concerns under healthcare data regulations.
- Solution: Implemented end-to-end encryption, consent-based data sharing, and time-limited access tokens for family tracking links.
- Learning: Privacy by design is non-negotiable in healthcare applications — it must be built in, not bolted on.

Scalability During Mass Emergency Events

- Challenge: During natural disasters or mass casualty events, the system must handle exponential load spikes.
- Solution: Designed for cloud-native auto-scaling with priority queue management for triage-level emergency requests.
- Learning: Scalability planning for worst-case scenarios is as important as designing for normal operations.

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